

A Socio-Technical Approach to Designing Digital Health Tools for Older Caregivers: Insights from a Case Study of the Next-of-Kin Compass

Thesis Report

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Popular Science Summary

Sweden’s population is ageing, putting pressure on the healthcare system. With not enough staff and more people needing care, hospitals and clinics are using more digital tools like apps and online services. These let people manage appointments, keep track of medications, and contact their doctor from home.

For many people, these tools work well. But for older family caregivers, such as spouses, adult children, and partners caring for someone with cancer, they often do not. These caregivers are already handling medications, appointments, and emotional stress for someone else, while also facing their own challenges like poor eyesight, memory issues, or not being used to new technology. A confusing app does not make things easier for them. It makes things harder.

We wanted to understand why digital tools so often fail this group, and what would need to change for them to actually help. We examined Närståendekompassen (Next-of-Kin Compass), a Swedish digital tool designed to support cancer caregivers, by evaluating its usability, conducting a workshop with researchers in this field, and interviewing caregiving experts and one older caregiver with lived experience.

What we found was consistent. Older caregivers do not need impressive technology. They need tools that are calm, predictable, and quick to use, tools that work with their circumstances rather than against them. From these findings, we developed concrete design guidelines and applied them directly to the Next-of-Kin Compass, showing how targeted changes to labels, content, accessibility features, and onboarding can make a meaningful difference.

The broader point matters beyond this one tool. As more healthcare moves online, designing for the realities of older caregivers' lives is not optional; it is essential.

Abstract

Sweden's ageing population and shrinking medical workforce are placing pressure on the healthcare system, leading to greater dependence on digital healthcare. Informal caregivers, frequently older adults, are filling gaps in formal care without adequate support for their well-being. While digital tools have the potential to support them, these often fail to meet the specific needs of older caregivers, who juggle age-related challenges alongside demanding caregiving duties. Research has often treated ageing and caregiving issues separately, resulting in a limited understanding of their combined impact on digital health design.

This thesis explores how the challenges of ageing and caregiving can inform the design of digital health tools for older caregivers, focusing on the development of design guidelines that can be applied in real-world contexts. To address this, the study adopts a single case study approach focused on the Swedish caregiver support tool Närståendekompassen (Next-of-Kin Compass). Data were collected through heuristic evaluation, a reflection workshop, semi-structured interviews, and iterative prototype redesign. Data were analyzed using thematic analysis and interpreted through the Senior Technology Acceptance Model (STAM) and the Non-adoption, Abandonment, Scale-up, Spread, and Sustainability (NASSS) framework.

The findings show that older caregivers experience overlapping cognitive, emotional, physical, and time-related pressures that influence digital health engagement. In response, the study identified design guidelines centered on clarity, simplified navigation, flexibility, trust, and supportive guidance, which were applied in the redesign of the Next-of-Kin Compass. This thesis enhances digital health and Human-Computer Interaction research by identifying older caregivers as a unique user group with specific design needs.

Keywords: Older Adults, Ageing, Caregiving, Human-Computer interaction, Design Guidelines, Digital Health

Declaration of AI Usage

During thesis preparation, we used artificial intelligence tools in specific, limited ways, described below:

- Grammarly for grammar checking, spelling correction, and surface-level language refinement.
- ChatGPT (GPT-4) and Google Gemini were occasionally used to explore alternative phrasing, organisational structures, and clarity of written analytical sections already developed by the authors. All coding decisions, thematic interpretations, and analytical conclusions were independently conducted and validated by the authors.
- Microsoft Copilot for formatting and layout assistance in presentation materials.

We critically evaluated all AI-generated suggestions and either substantially revised or discarded them. AI tools were used only in a supportive capacity for writing and structural refinement and did not replace the authors' own analytical judgement, interpretation of empirical data, or development of conclusions.

All research questions, methodological decisions, data collection, coding, thematic interpretation, and final arguments represent our original work. We independently identified, verified, and reviewed all references and citations using primary sources.

We recognise that AI language models may produce inaccuracies. Therefore, we critically evaluated all AI-generated output before use. We alone are responsible for the accuracy, originality, and academic integrity of this thesis.

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Swedish Terminology

Närståendekompassen	“Next-of-Kin Compass” – digital support platform for family caregivers of cancer patients within the Swedish healthcare system.
Socialstyrelsen	National Board of Health and Welfare – government agency responsible for regulations, guidelines, and national health/social care statistics.
Regeringskansliet	Government Offices of Sweden – prepares and coordinates government policy, including digitalisation strategies.
Vård och omsorgsanalys	Swedish Agency for Health and Care Services Analysis – evaluates efficiency, quality, and accessibility of health and care services.
Statistiska centralbyrån (SCB)	Statistics Sweden – national statistical authority.
Fast omsorgskontakt	Permanent care contact – a licensed assistant nurse (*undersköterska*) serving as a primary point of contact for older adults receiving home care.
Undersköterska	Licensed assistant nurse – key professional role in Swedish elderly care.
1177 Vårdguiden	Sweden’s national online healthcare portal and telephone advice line.
NKA – Nationellt kompetenscentrum för anhöriga	Swedish National Competence Centre for Relatives – provides knowledge, training, and support for informal carers.
Lag om tillgänglighet till digital offentlig service (SFS 2018:1937)	Act on Accessibility to Digital Public Services – Sweden’s implementation of the EU Web Accessibility Directive.

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1. Introduction

Sweden is experiencing a demographic shift characterized by an ageing population and a widening care gap. Currently, around 20% of the Swedish population is 65 or older (World Bank, 2025), and this figure is projected to rise to 23% by 2040 (Swedish Institute, 2020). Importantly, older adults are one of the main users of healthcare services, as they often live with multiple health conditions and long-term care needs (Åberg et al., 2024). Although demand for care is growing, the welfare sector is struggling to recruit and retain healthcare personnel (Penje & Berlina, 2021). This places sustained pressure on the national health and social care systems (Åberg et al., 2024). In response to these pressures, national strategies have emphasized digitalization as a means of promoting the long-term sustainability, accessibility, and efficiency of healthcare (Regeringskansliet, 2023).

As a result, digital health applications have become increasingly important for facilitating access to care. However, the success of digitalization does not solely depend on the availability of technology. It is largely shaped by users' ability and willingness to engage with it. This presents a particular challenge for older adults, who often encounter barriers that affect their use of technology, such as changes in visual acuity and hand-motor coordination (Liu et al., 2021). Consequently, despite significant investments in healthcare digitalization, adoption of digital health tools remains limited among older adults in Sweden (Åberg et al., 2024).

The continued gap between the potential of digital health technologies and their actual use reflects broader challenges within the Swedish healthcare system. Although digitalization is often presented as a way to address growing healthcare demands, significant deficits remain in the provision of formal care (Socialstyrelsen, 2025). These deficits are increasingly compensated for by informal caregivers who provide practical and emotional support to relatives and friends with health needs (Kirvalidze, 2026).

Importantly, many informal caregivers are themselves older adults (Kirvalidze, 2026). This places them in a unique position within the healthcare system, as they may simultaneously experience age-related challenges while carrying substantial caregiving responsibilities. Informal caregiving often involves coordinating care, managing information, providing emotional support, and assisting with everyday activities. These responsibilities can negatively affect caregivers' physical and psychological well-being, particularly when support structures are insufficient (Samuelsson et al., 2021; Glasdam et al., 2025; Kirvalidze et al., 2025). Older caregivers may therefore experience a dual vulnerability arising from both the ageing process and the demands of caregiving (Ahmad et al., 2023; Kirvalidze et al., 2025). An example of the older caregiver and their challenges can be observed in Figure 1 through an illustrative user persona.

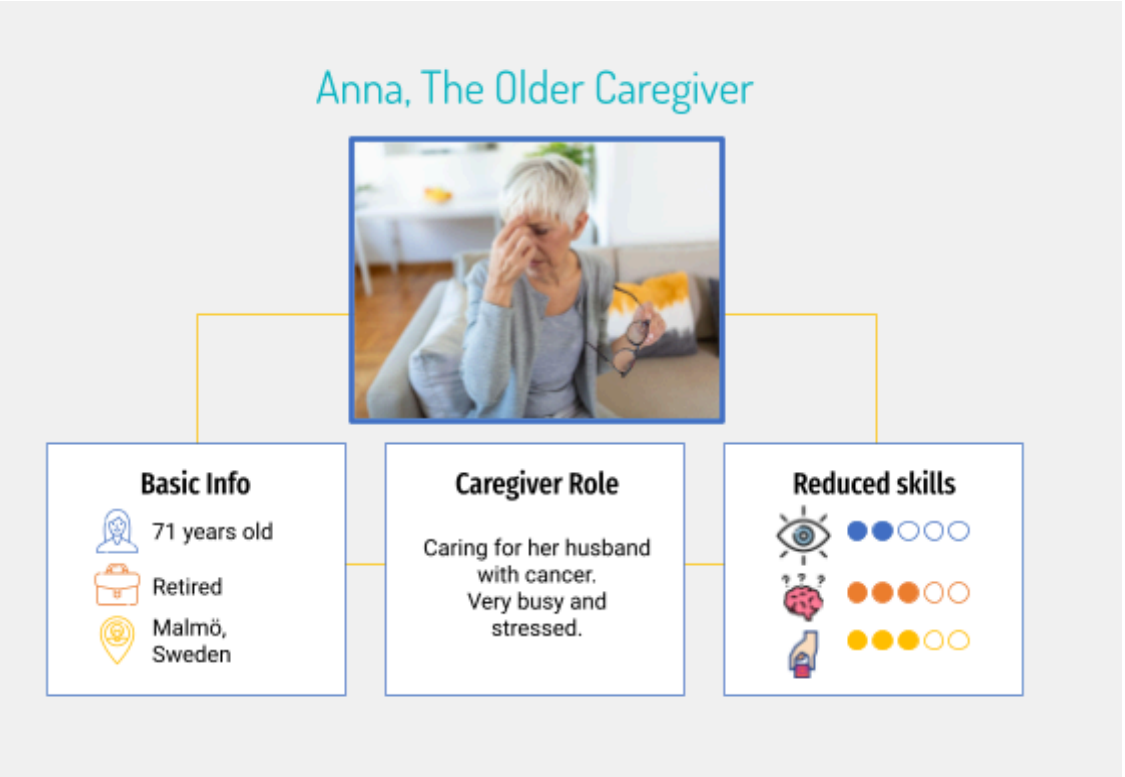


Figure 1. The Older Caregiver. This figure shows a user persona for the older caregiver, showing their basic information, their role as a caregiver, and their age-related reduced skills such as vision, cognitive and motor limitations.

As both care providers and recipients, older caregivers might benefit from digital health interventions designed to support them, considering the fragmented care they receive from formal services. However, the older caregiver's dual vulnerability raises important questions for digital health design regarding whether existing technologies adequately support individuals navigating both the challenges of ageing and the demands of caregiving simultaneously.

1.1 Problem Statement and Motivation

Digital tools designed to support caregivers have demonstrated potential to reduce caregiver burden by improving access to information, facilitating care coordination, and helping users obtain relevant support services (Ancona et al., 2025; Zhai et al., 2023). However, existing caregiver technologies might not adequately address the needs of older caregivers. Research suggests that many digital health tools remain insufficiently adapted to age-related and accessibility requirements (Liu et al., 2021; Åberg et al., 2024; Denecke et al., 2025; Franklin & Sahiti, 2018).

The ageing process involves a complex interaction of sensory, motor, and cognitive changes that affect how individuals interact with digital technologies (Liu et al., 2021). For older caregivers, these age-related factors intersect with the practical and emotional demands of caregiving. As a result, their experiences with digital health technologies are shaped not only by ageing but also by the context and responsibilities of caregiving (Tan et al., 2025).

While previous research has explored technology use among older adults and digital support for caregivers separately, limited attention has been given to understanding the specific challenges that emerge at the intersection of ageing and caregiving. Consequently, there remains a need to identify these challenges and examine how they can inform the design of digital health technologies intended for older family caregivers. Addressing this gap is important for ensuring that digital health innovations are not only technologically functional but also accessible, meaningful, and supportive for this population.

To address this gap, this thesis examines how ageing- and caregiving-related challenges influence the design of digital health tools for older family caregivers. The study focuses specifically on older family caregivers rather than older adults more generally, as this group experiences the combined effects of ageing and caregiving responsibilities.

The research is conducted as a single case study of the Next-of-Kin Compass tool (Närståendekompassen), a digital support platform developed for family caregivers of cancer patients within the Swedish healthcare system. This tool was selected because it represents an initiative specifically designed to support caregivers. However, the design process of this tool has not explicitly focused on the needs and contextual challenges of older caregivers (Samuelsson, 2025). Thus, this case allows exploring ageing- and caregiving-related challenges and how they can inform design choices in a real-world case. The study therefore seeks both to generate broader design knowledge and to demonstrate its practical application. While the case study is grounded in the specific context of the Next-of-Kin Compass, the design guidelines developed are intended to have broader applicability to digital health tools for older caregivers.

1.2 Purpose, Aim, and Objectives

The purpose of this thesis is to examine how ageing- and caregiving-related challenges can inform the design of digital health tools that better support older family caregivers.

The study aims to identify older family caregivers challenges that shape digital health engagement, develop context-sensitive design guidelines, and explore their application in the Next-of-Kin Compass tool.

To achieve this aim, the study addresses the following objectives:

1. Identify ageing- and caregiving-related challenges that influence older family caregivers' use of digital health tools.
2. Develop design guidelines that address these challenges.
3. Explore how these design guidelines can be applied in the redesign of the Next-of-Kin Compass tool.

1.3 Research Questions

The first two questions address the broader design challenge of supporting older family caregivers through digital health tools, while the third examines how this knowledge can be applied within a specific real-world case.

RQ1: What ageing- and caregiving-related challenges shape older family caregivers' use of digital health tools?

RQ2: How do these challenges inform design guidelines for digital health tools so they can be supportive for older family caregivers?

RQ3: How can these design guidelines be applied in the redesign of the Next-of-Kin Compass tool?

1.4 Target Audience and Limitation

This thesis is intended for designers, developers, and researchers in Human-Computer Interaction (HCI), digital health, and ageing studies. The findings are also relevant to healthcare organisations and practitioners involved in the development or implementation of digital support tools for caregivers.

This study focuses on older family caregivers of cancer patients and employs the Next-of-Kin Compass as a single case study. Consequently, the findings are not intended to be universally generalizable across all caregiving contexts or digital health technologies. Nevertheless, the case study approach enables an in-depth examination of how ageing- and caregiving-related challenges influence digital health design.

While grounded in a specific context, the resulting insights and design guidelines may be transferable to the development and adaptation of digital health tools for other caregiving populations and older adults more broadly. In this way, the study contributes to broader discussions on context-sensitive design in digital health for ageing and caregiving populations.

1.5 Thesis Structure

The remainder of this thesis is organised as follows:

- Section 2 reviews literature on ageing, caregiving, healthcare digitalization and digital health design for older adults and introduces the theoretical frameworks used.
- Section 3 outlines the research methodology employed in this study, including the case study approach, the case context, data collection methods, prototype redesign process, and data analysis procedures.
- Section 4 presents the results derived from the heuristic evaluation, workshop activities, prototype redesign, and stakeholder interviews.
- Section 5 analyzes the findings through theoretical frameworks.
- Section 6 discusses the study's broader implications for digital health design and caregiving support, while reflecting on the thesis limitations.
- Section 7 concludes the thesis by summarising the main contributions of the study and outlining directions for future research.

2. Background

This section outlines the demographic, technological, and regulatory context shaping older informal caregivers' engagement with digital health in Sweden. It reviews relevant digital tools and key analytical frameworks, establishing the foundation for understanding current challenges and gaps in the literature.

2.1 Older Caregivers in Sweden

Demographic and structural changes in Sweden are reshaping both the need for and provision of informal care. The following subsections address population ageing, healthcare digitalization efforts, and the growing dependence on older adults as informal caregivers.

2.1.1 Ageing Population Challenges

Sweden, like other Nordic countries, is experiencing a demographic shift characterized by an ageing population and a widening care gap. Older adults constitute one of the main user groups of health and social services, as age-related multimorbidity and chronic conditions necessitate high-frequency interaction with formal care systems (Åberg et al., 2024). Currently, around 20% of the Swedish population is 65 or older (World Bank, 2025), and this figure is projected to rise to 23% by 2040 (Swedish Institute, 2020). The most acute growth is seen in the oldest age brackets, as Socialstyrelsen (2025) reports that the population aged 80 and older will increase by 38% by 2033 (Statistiska centralbyrån, 2024). This demographic pressure is not evenly spread out across the country. Younger people are moving to cities, which means that some rural areas have more than 35% of their population over the age of 65 (Dzhavatova et al., 2025; Backman & Karlsson, 2023).

This demographic change is occurring simultaneously with a critical workforce contraction. Socialstyrelsen (2025) highlights a declining number of both assistant and registered nurses within elderly care. Penje & Berlina (2021) note that this challenge is shared across Nordic welfare states: demand for care is growing while the welfare sector struggles to recruit and retain staff. Ultimately, the combination of rising demand and a shrinking professional workforce places sustained pressure on the national health and social care systems (Åberg et al., 2024). As home care increases, the Swedish government introduced a permanent care contact ("fast omsorgskontakt"), a licensed assistant nurse ("undersköterska") who serves as a primary point of contact for an older adult receiving care at home. Even though the role is intended to ensure continuity of care, access to this resource remains inconsistent (Socialstyrelsen, 2025).

2.1.2 Healthcare Digitalization

Meanwhile, Sweden has pursued healthcare digitalization as part of a broader digitalization strategy (Åberg et al., 2024) to alleviate pressure on the welfare system. With a projected shortfall of 91,000 healthcare employees and a SEK 75 billion funding gap by 2040, the state has framed digitalization as a non-negotiable prerequisite for maintaining public commitment to care (Vård och omsorgsanalys, 2024). This transformation, which accelerated significantly following the COVID-19 pandemic (Åberg et al., 2024), positions digital health as a key mechanism to improve access to care, increase efficiency, and manage resource constraints (Åberg et al., 2024) by introducing less staff-intensive working methods (Vård och omsorgsanalys, 2024).

Even so, despite the increasing digitalization of the healthcare system, adoption of digital health tools remains limited among older adults in Sweden (Åberg et al., 2024). While some evidence suggests moderate-to-high levels of eHealth literacy in this demographic, increasing age continues to be a strong predictor of lower adoption rates (Ghazi et al., 2023). This reflects a broader pattern: reforms like digitalization aim to help bridge the care gap as the population continues to age, but there are still deficits in the formal healthcare system (Socialstyrelsen, 2025; Kirvalidze, 2026).

2.1.3 The Rise of Older Informal Caregivers

Due to the shortcomings in the formal healthcare system, care is predominantly maintained by individuals who unofficially assume the role of caregivers to address the deficiencies (Socialstyrelsen, 2025; Kirvalidze, 2026). This role often reflects "morally driven labor": work motivated by deep personal duty and relational bonds rather than professional choice (Glasdam et al., 2025). Because this labor is informal, it often remains invisible within national policy frameworks (Glasdam et al., 2025).

Importantly, a significant amount of older adults act as informal caregivers. In Sweden, approximately 19.2% of adults over the age of 60 occupy this role (Kirvalidze, 2026), frequently delivering high-intensity support to patients (Kirvalidze et al., 2025). This dual position as both potential care recipients and providers makes these individuals a particularly complex and vulnerable group. This vulnerability is intensified by the specific nature of the caregiving role, which extends beyond physical assistance to encompass a multifaceted and often overwhelming set of responsibilities.

The caregiving roles assumed by older adults are not only practical but multidimensional, encompassing emotional, informational, and coordination-related tasks (Samuelsson et al., 2021; Glasdam et al., 2025). In addition, the sustained demands often lead the caregiver to prioritize the patient's needs at the expense of their own well-being (Glasdam et al., 2025). Consequently, caregivers frequently experience substantial burden and associated health risks (Kirvalidze et al., 2025).

In turn, caregivers often have needs that are not adequately met by current support systems (Samuelsson et al., 2021), which directly impacts their quality of life (Wang et al., 2021). Importantly, older caregivers face a compounded susceptibility because of their own age-related vulnerabilities (Ahmad et al., 2023; Kirvalidze et al., 2025). In fact, nearly 10% of older caregivers in Sweden receive care themselves (Kirvalidze, 2026).

Ultimately, as both care providers and recipients, older caregivers might benefit from digital health interventions designed to support them, considering the fragmented care they receive from formal services. This necessitates a closer examination of the digital tools currently available for supporting them and their adoption.

2.2 Digital Tools for Older Caregivers

A growing body of research has examined the potential of digital tools to support informal caregivers, yet significant gaps remain in how well these tools serve older users specifically. This section reviews evidence on caregiver-focused digital interventions before examining the broader literature on older adults' interactions with digital health technologies.

2.2.1 Support for Caregivers

Digital tools designed to assist caregivers have shown potential to mitigate caregiver burden. In the European context, as described by Ancona et al. (2025), these solutions typically cluster into four functional domains: (i) psychophysical well-being (e.g., mindfulness and virtual support groups), (ii) management of physical distress (e.g., dietary and exercise platforms), (iii) remote monitoring of vitals and symptoms, and (iv) empowerment through self-efficacy tools such as educational resources.

A systematic review by Zhai et al. (2023) confirms that digital interventions can robustly enhance caregivers' psychological health, self-efficacy, and problem-coping abilities. By providing real-time, up-to-date information and interactive communication platforms, these tools often compensate for the inherent limitations and inconsistencies of traditional human-led healthcare interactions. Likewise, a 2026 scoping review of 110 studies found positive effects across burden, well-being, competence, quality of life, relationships, and care coordination (Nittas et al., 2026).

However, high technical effectiveness in controlled settings does not guarantee benefit for the broader caregiver population. While Ancona et al. (2025) report that interventions for psychophysical well-being reach up to an 80% effectiveness rate in studies, the generalizability of these results remains limited by a disproportionate representation of younger, more digitally literate participants and a lack of focus on specific cultural and situational contexts of informal caregivers. Even more, Nittas et al. (2026) highlight accessibility as a significant barrier for the generalizability of positive effects.

Consequently, the unique functional capabilities and preferences of elderly users are routinely overlooked during tools' development (Liu et al., 2021). This results in a landscape where, despite older users being one of the primary users of health and social care services, digital health solutions remain insufficiently adapted to their specific needs (Åberg et al., 2024; Denecke et al., 2025; Franklin & Sahiti, 2018).

This design mismatch is evidenced when looking at older spousal caregivers, a demographic whose specific context is underserved by current applications. Empirical data from Raj et al. (2022) indicates that older spousal caregivers are significantly less likely to engage with mobile health (mHealth) technologies than younger caregivers, driven by lower digital literacy and a perceived lack of utility of the tools. Importantly, older spousal caregivers often manage their own age-related health declines alongside those of the patient (Raj et al., 2022), creating a complex context that generalized designs fail to engage.

Given this, bridging the gap between technological potential and actual usage requires a move toward more demographically sensitive design (Zhai et al., 2023). This is particularly critical for older caregivers, whose caregiving responsibilities are layered onto age-related physical, cognitive, and digital barriers that already complicate digital health adoption among older adults more broadly. Understanding why digital tools can fail to adequately support this group, therefore, requires examining the broader literature on older adults' interactions with digital health technologies.

2.2.2 Older Adults in Digital Health

Within the broader older adult population, a persistent gap exists between interest in digital healthcare services and actual adoption (Nymberg et al., 2019; Berner et al., 2022). Multiple interacting barriers shape this gap (Åberg et al., 2024).

First, there are multiple age-related factors that influence adoption. For example, the ageing process negatively affects visual acuity, hearing, and hand-motor coordination (Liu et al., 2021). Specific challenges include reduced depth perception, low contrast sensitivity, and tremors. These make interacting with small screens and dense layouts exceptionally difficult (Amouzadeh et al., 2025). In addition, older adults face declines in information processing capacity and spatial cognition (Liu et al., 2021). High cognitive load is a major barrier, often caused by complex menu hierarchies. However, a major gap exists in the literature, as most studies infer this cognitive burden indirectly through satisfaction scores rather than using direct task-based or physiological measurements (Chirayus & Nanthaamornphong, 2019).

Moreover, many adults exhibit an ambivalence toward digital health (Nymberg et al., 2019), rooted in concerns over trust and a lack of perceived usefulness, as seniors tend to prioritize apps they consider essential to their daily needs (Sze et al., 2023; Liu et al., 2021). Similarly, an obstacle of adoption for some seniors is a general belief that technology is unnecessary in certain areas of life (Tajudeen et al., 2022).

Additionally, adoption is hindered by a decline in digital confidence (Nymberg et al., 2019) and an increase in technology-related anxiety (Berner et al., 2022), both of which correlate with advancing age (Åberg et al., 2024). Moreover, the rapid pace of digital innovation makes it difficult for older adults to keep up (Åberg et al., 2024). The challenge is exacerbated by a systemic trend where the burden of acquiring digital competence is placed on individuals rather than supported through institutional or social infrastructure, leading to generally lower digital literacy levels (Nymberg et al., 2019; Åberg et al., 2024).

Because of these persistent barriers, older users rarely access digital health individually. Informal support networks, such as family members, friends, or neighbors (Rasi et al., 2020), often mediate this process. Importantly, while this kind of mediation makes it easier to use these tools in the short term, it also limits how freely older users can use them (Åberg et al., 2024).

The literature has identified several requirements to address usability barriers for older users, taking into account the difficulties that they encounter. These considerations include, but are not limited to:

- **Interface Elements:** Recommendations include large sans-serif fonts (at least 20–30 points), high-contrast color schemes (minimum 4.5:1 ratio), and touch targets of at least 15mm in diameter (Morey et al., 2019).
- **Navigation and Layout:** Research favors flat navigation structures where main functions are visible on the home screen to minimize steps (Liu et al., 2021). Additionally, hiding nonessential information by default, avoiding ambiguity, and providing clear, labeled buttons can reduce cognitive load (Mattick et al., 2024).
- **Support and Learning:** Because exploratory learning is often ineffective for this demographic, research favors step-by-step guidance, and video tutorials are encouraged. Having a human in the loop of support has also been recommended, such as through technical coaching (Gomez-Hernandez et al., 2023; De Santis et al., 2023).

While these evidence-based design standards provide a baseline for age-friendly interaction, they often assume a solitary user in a stable environment. This highlights a significant limitation in current digital health, which is the failure to move from

ageing-friendly interfaces to tools that acknowledge the multidimensional reality of caregiving and ageing.

2.3 Theoretical and Regulatory Frameworks

The literature reviewed so far demonstrates that there is a mismatch between design assumptions and the challenges of the older caregiver. Although existing research offers valuable recommendations for improving accessibility among older adults, many solutions aimed at supporting caregivers still rely on generalized user assumptions and insufficiently reflect the realities of different user contexts, limiting their positive impact. Because older caregivers often manage emotional strain, coordination work, time pressure, and their own health-related challenges in parallel, digital health use is embedded within ongoing caregiving practices rather than isolated interactions with technology.

Thus, barriers are not reducible to individual deficits or interface shortcomings. Instead, they emerge from misalignments between technological assumptions and the social contexts in which systems are used. Recent digital health research supports this perspective, showing that adoption and sustained use are shaped by interacting factors across individual and contextual levels (Kronlid et al., 2024; Hamasaki et al., 2025). Digital health engagement is therefore best understood as a socio-technical process rather than a discrete user–interface interaction.

Therefore, understanding the adoption and use of digital health tools by older caregivers requires analytical frameworks that can account for both individual capacity constraints and broader system-level complexity. A range of theoretical models and regulatory frameworks have been developed in this space, each offering different analytical affordances. This section provides an overview of key frameworks that are relevant to studying digital health engagement in caregiving contexts.

2.3.1 Technology Acceptance Models

The Technology Acceptance Model (TAM; Davis, 1989) and its successors, the Unified Theory of Acceptance and Use of Technology (UTAUT; Venkatesh et al., 2003) and its consumer-oriented extension UTAUT2 (Venkatesh et al., 2012), have been widely used to explain why individuals adopt or reject digital tools. Both TAM and UTAUT center on perceived usefulness and ease of use as primary predictors of adoption intention, while UTAUT2 extended this framework to voluntary consumer contexts by incorporating additional predictors such as hedonic motivation and habit.

While influential, these generalist models have been critiqued for insufficiently accounting for the characteristics of specific user groups, particularly older adults. This limitation is especially relevant in the context of mHealth and caregiver digital tools, where adoption is voluntary and shaped by whether users find the technology meaningful and essential to their daily lives (Sze et al., 2023; Liu et al., 2021), precisely the dimensions UTAUT2 sought to capture.

In response to the limitations of generalist models, Chen & Chan (2014) developed the Senior Technology Acceptance Model (STAM), which integrates age-related variables that standard acceptance models had not adequately addressed. Unlike TAM and UTAUT, STAM explicitly incorporates gerontechnology anxiety, cognitive self-efficacy, functional ability, and health status as antecedent variables that shape the baseline capacity available for engaging with digital systems. In doing so, it reconceptualizes the preconditions for engagement rather than simply adding variables alongside standard

TAM constructs. This positions STAM as a useful framework for understanding technology adoption among older adults, including those in caregiving contexts where older adults must navigate digital tools while simultaneously managing age-related limitations and sustained caregiving demands.

2.3.2 System-Level Complexity Frameworks

Beyond individual-level acceptance, digital health adoption is also shaped by the organizational, technological, and institutional contexts in which tools are introduced. The Non-adoption, Abandonment, Scale-up, Spread, and Sustainability (NASSS) framework, developed by Greenhalgh et al. (2017), addresses this system level. Rather than treating adoption as the outcome of individual decisions, NASSS conceptualizes it as the product of interacting technological, organizational, and contextual factors across multiple domains.

This perspective is particularly relevant in digital health, where technologies are often introduced with implicit assumptions that innovation will automatically translate into improved care efficiency and outcomes (Greenhalgh et al., 2017). NASSS challenges this assumption by showing that adoption and sustained use are contingent on the interaction between multiple layers of complexity, across users, technologies, organizations, and broader institutional environments. Recent digital health research supports this perspective, demonstrating that adoption and sustained use are shaped by interacting factors at both individual and contextual levels (Kronlid et al., 2024; Hamasaki et al., 2025).

Other socio-technical frameworks, such as Normalization Process Theory (NPT; May & Finch, 2009), have similarly been applied to understand how digital health interventions become embedded or fail to become integrated in care practices. NPT focuses on the work people do to implement and sustain complex interventions, examining coherence, cognitive participation, collective action, and reflexive monitoring. Together, these frameworks reflect a broader move in digital health research toward socio-technical perspectives that treat adoption as a process shaped by social contexts rather than solely by individual attitudes.

2.3.3 Accessibility Legislation and Design Standards

Alongside theoretical frameworks, regulatory and legal frameworks increasingly shape how digital health tools are designed and evaluated. In Europe, the EU Web Accessibility Directive (Directive 2016/2102/EU, 2016) mandates that public sector digital services meet accessibility standards, primarily defined by the Web Content Accessibility Guidelines (WCAG 2.1; W3C, 2018).

More recently, the European Accessibility Act (Directive 2019/882/EU, 2019), which entered into force in April 2019, extends accessibility requirements to a broader range of products and services in the private sector, including digital services and consumer technology. In Sweden, these directives are transposed into national law through the Act on Accessibility to Digital Public Services (SFS 2018:1937, 2018).

WCAG 2.1 organizes accessibility criteria around four principles: that content must be perceivable, operable, understandable, and robust (POUR; W3C, 2018). These principles translate into concrete design requirements such as minimum contrast ratios, keyboard navigability, and text alternatives for non-text content that closely align with the usability recommendations identified in the older adult digital health literature. However, a notable gap persists between legal accessibility compliance and genuinely

age-inclusive or caregiver-sensitive design. Compliance with accessibility standards provides a necessary but insufficient foundation; it does not, on its own, address the cognitive, emotional, and contextual demands of older caregivers during digital interaction.

Many of the usability barriers identified among older adults, including reduced visual acuity, impaired motor control, and increased cognitive demands, correspond directly to accessibility challenges addressed within standards such as WCAG. This creates an important overlap between accessibility compliance and the design recommendations emerging from research on older adults' digital health use. The intersection of accessibility law, design standards, and theoretical frameworks therefore offers a multidimensional basis for evaluating and developing digital health tools, one that encompasses both minimum legal requirements and the more nuanced demands of specific user populations.

Together, these frameworks illuminate different aspects of digital health engagement. Technology acceptance models primarily focus on individual perceptions and capabilities, socio-technical frameworks emphasize interactions between users and broader systems, and accessibility standards establish minimum requirements for equitable access. Considered collectively, they suggest that successful digital health adoption depends not only on user motivation or technological quality but also on how technologies align with the capabilities, contexts, and needs of intended users.

2.4 Limitations of Existing Knowledge

The literature reviewed throughout this section demonstrates that older caregivers occupy a complex position as users within digital health. Existing research has established that older adults face sensory, cognitive, emotional, and digital literacy barriers when interacting with health technologies (Åberg et al., 2024; Liu et al., 2021), while caregiving research separately shows that informal caregivers experience multidimensional burden associated with coordination work, emotional strain, and sustained responsibilities (Kirvalidze et al., 2025; Glasdam et al., 2025). However, these bodies of literature largely remain analytically separated.

Meanwhile, digital health research insufficiently addresses how ageing-related capacity constraints interact with the practical and emotional demands of caregiving during technology use. Most age-friendly design research focuses primarily on interface accessibility and usability optimization for older adults in general. Simultaneously, caregiver-focused digital interventions show positive psychological and functional outcomes but fail to adequately account for the age-related limitations of older caregivers, limiting their benefits for the older caregiver. Consequently, older caregivers are often treated either as generic older adults or as generic caregivers, rather than as a distinct socio-technical user group shaped by the intersection of both conditions.

This separation creates several important limitations in existing knowledge. First, much of the literature emphasizes individual usability barriers while underexamining how healthcare system fragmentation, the complexity of caregiving, and informal care responsibilities shape digital health engagement in practice. Second, although studies identify barriers such as cognitive overload, low digital confidence, and technology anxiety, these factors are often examined independently rather than as cumulative and interacting pressures within caregiving environments. Third, existing design recommendations remain largely focused on interactions, emphasizing interface clarity, navigation simplicity, and accessibility standards, while giving comparatively limited

attention to how caregiving contexts alter users' available cognitive, emotional, and temporal resources during digital interaction.

Furthermore, research addressing older caregivers as a distinct user group within digital health remains limited, which is particularly significant in Nordic and Swedish contexts characterized by demographic ageing and increasing reliance on informal care. Existing studies rarely examine how age-related constraints and caregiving responsibilities interact to shape digital health engagement and design needs in practice.

Accordingly, a significant research gap exists in understanding how ageing and caregiving challenges interact to shape requirements for digital health design. Existing literature suggests that understanding digital health engagement among older caregivers may benefit from approaches that move beyond generalized models of older adults or caregivers and instead consider the interaction between individual capacities and broader socio-technical contexts.

3. Methodology

This section outlines the methodological framework and design choices utilized to investigate the research questions in the study. To provide a comprehensive understanding of the research process, the section is structured into seven primary sections. First, the overarching theoretical foundation of case study methodology is established, followed by a detailed description of the specific empirical case: the Next-of-Kin Compass. Next, the general study design and its consecutive research stages are outlined. The section then details the specific data collection methods and thematic data analysis procedures employed. Finally, the ethical considerations guiding the study are discussed, alongside a reflexive evaluation of alternative methods that were considered but ultimately dismissed.

3.1 Theory: Case Study Methodology

Case study research is a methodological approach that enables exploration of phenomena within their real-world contexts, particularly when boundaries between phenomenon and context are not clearly separable (Greenhalgh, 2025; Yin, 2018). Particularly, qualitative case studies prioritise depth, contextual richness, and interpretive understanding, rather than universal generalisability, statistical validity, or reproducibility (Greenhalgh, 2025; Noble & Smith, 2025; Yin, 2018). Even so, sufficient contextual description can support transferability to other comparable settings (Greenhalgh, 2025; Lazar et al., 2017).

Case study research can incorporate multiple data collection methods. Data triangulation is commonly used to strengthen construct validity by examining the phenomenon through different perspectives and data types, such as interviews or artefacts (Noble & Smith, 2025). This approach reduces dependence on a single interpretive lens and enables convergence or divergence across data sources to be systematically analysed. In addition, a case study can involve iteration between data collection, analysis, and interpretation, particularly within qualitative and design-oriented research contexts (Yin, 2018). In such approaches, emerging findings can continuously inform subsequent stages. Finally, case study designs can be

categorised as multiple-case or single-case studies; the latter enables in-depth examination of one bounded case (Yin, 2018).

3.2 Case Description: Next-of-Kin Compass

The case examined in this study is the Next-of-Kin Compass (Närståendekompassen), a digital support tool currently under development within the FamCASP research initiative at Malmö University's Sustainable Digitalisation Research Centre.

The Next-of-Kin Compass is intended to support family caregivers of cancer patients in Sweden. The main purpose of the tool is to help caregivers understand their needs better. Thus, it helps users evaluate how well their needs are being met. In addition, the tool directs users toward relevant informational and support resources to provide insight into how to meet their needs better. A personal notes function also exists to allow users to record important dates and thoughts related to the illness trajectory or their own caregiving journey.

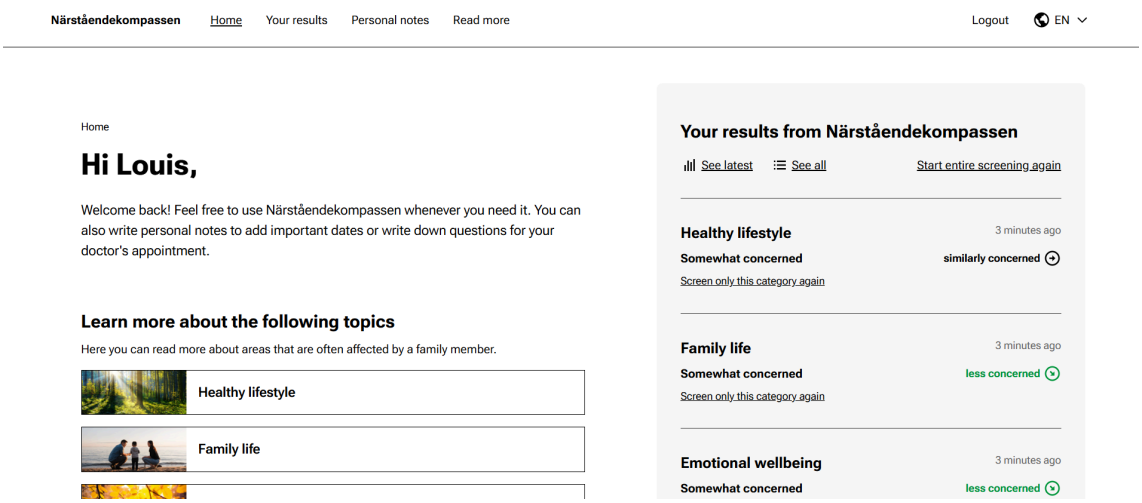


Figure 2. Main Interface. This view displays the homepage of the tool, where it shows previous results from a needs assessment to the right and information links to the left. In the top menu, the personal notes feature can be accessed.

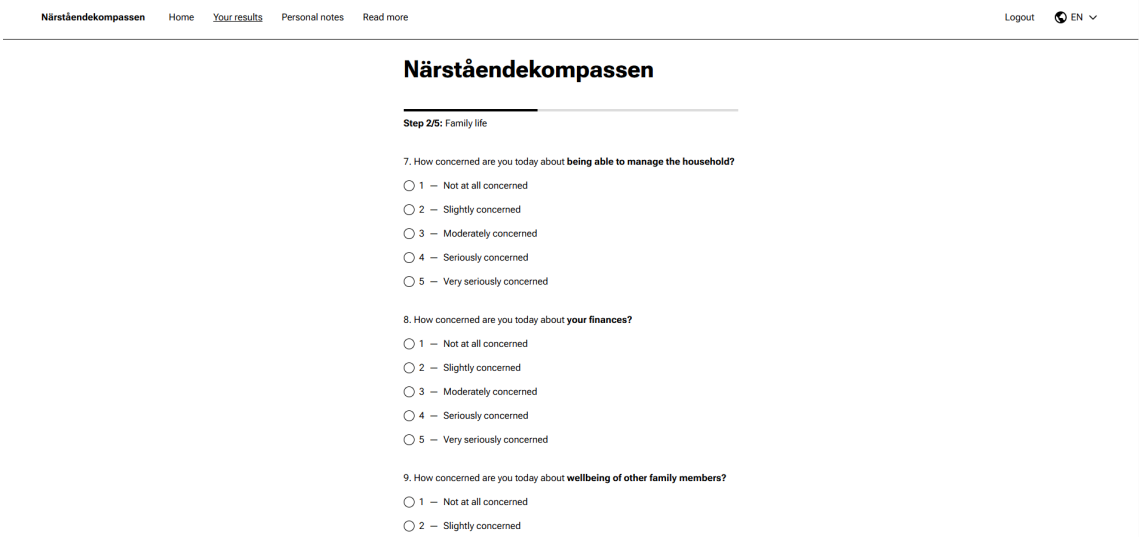


Figure 3. Assessment feature interface. This view displays the assessment feature of the tool, where users answer questions related to their needs throughout different steps or categories.

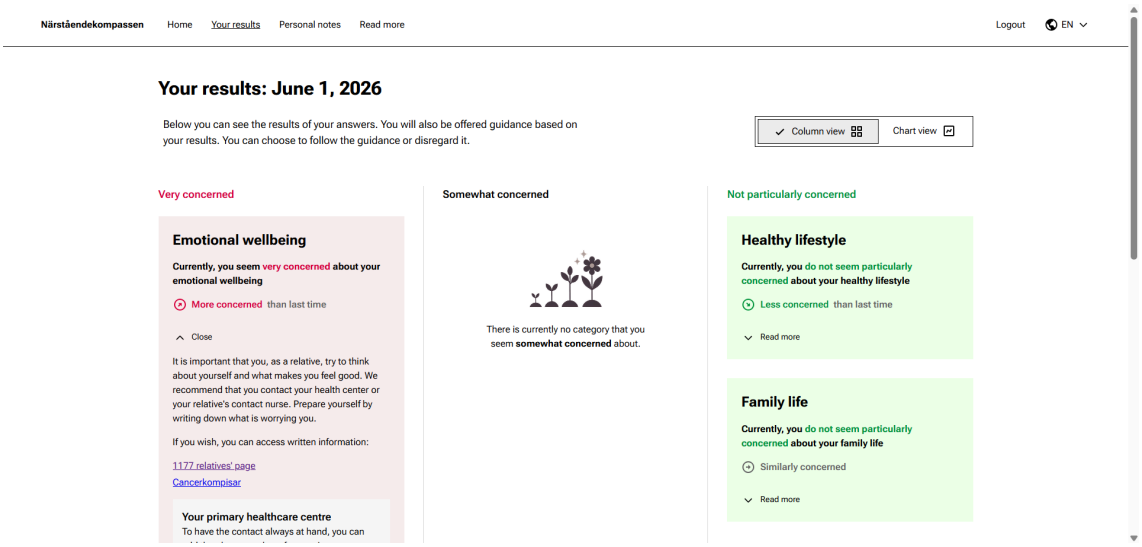


Figure 4. Results interface. This view displays one of the results obtained from the assessment of needs. Users can observe what categories they are very concerned, somewhat concerned, and not particularly concerned by. For example, emotional well-being is a category the user is very concerned about.

From a case study perspective, the Next-of-Kin Compass constitutes a bounded socio-technical case in which a digital artefact is situated within an ongoing development process and embedded within a real-world healthcare context. This makes it suitable for examining how usability considerations, caregiving and ageing demands, and design decisions interact during early-stage development.

In addition, caregiving for cancer patients is associated with significant emotional, informational, and organisational burden, including coordination with healthcare

services, monitoring patient needs, and navigating a complex system (Samuelsson, 2025). These demands make digital support tools particularly relevant but also highly sensitive to usability and accessibility constraints, especially for users with varying capacities and needs.

Although the Next-of-Kin Compass was developed to support caregivers broadly, its initial design process did not explicitly prioritise the specific needs of older caregivers, nor did it systematically account for how age-related changes in cognition, perception, motor function, or digital experience may interact with technology use. This absence creates a relevant analytical gap, making the case particularly suitable for investigating how ageing-related constraints and caregiving support can be integrated into digital health design processes.

Overall, the Next-of-Kin Compass case enables an in-depth exploration of how ageing, caregiving, and digital health design intersect within a real-world development setting rather than through a hypothetical or experimentally isolated system. This strengthens the contextual relevance of the findings and enables design choices to be explored within realistic constraints. Thus, the case enables the investigation of the study's research questions. Its ongoing development context allows the study to identify ageing- and caregiving-related challenges (RQ1) while also enabling iterative implementation and evaluation of design responses within the same system (RQ2 and RQ3).

3.3 General Study Design

The study will adopt a single-case study design centred on the Next-of-Kin Compass. The methodological design will be structured to address the research questions: RQ1, which focuses on ageing and caregiving challenges, RQ2, which asks how they inform the development of design guidelines for digital health tools, and RQ3, which focuses on how these guidelines can be applied within a real-world digital health tool.

To address RQ1, the study will prioritise methods that enable the identification and interpretation of caregiving and ageing-related challenges across multiple stakeholder perspectives. To address RQ2 and RQ3, the study will integrate research-through-design (RtD) principles, enabling iterative refinement of the Next-of-Kin Compass based on emerging findings and design guidelines. RtD treats design artefacts as objects that generate knowledge through development, use, and reflection (Zimmerman et al., 2007). Therefore, in this study, the Next-of-Kin Compass will function both as the bounded case and as a design artefact through which findings related to RQ1 are translated into design interventions addressing RQ2 and RQ3.

Thus, the study will combine multiple research and design activities and stakeholder perspectives (methodological triangulation). This strengthens the credibility and depth of interpretation by allowing findings to be examined across different forms of evidence rather than relying on a single interpretive viewpoint. In addition, findings, interpretations, and design decisions will continuously inform one another. Likewise, throughout the process, detailed documentation of methodological decisions, analytical interpretations, and redesign rationales will be maintained. This decision trail strengthens the transparency and traceability of how empirical findings inform subsequent design interventions and analytical conclusions. Even more, although the study will prioritise contextual understanding over statistical representativeness, providing detailed descriptions of the case, participants, and methodological process supports the transferability of findings to related digital health and caregiving contexts.

Considering that in caregiving contexts individual capacity constraints and system-level complexity interact to shape technology use in practice, this study will draw on two complementary frameworks to analyze digital health adoption among older caregivers through a socio-technical perspective. To achieve this, the study will analyse data through the interpretative lenses of the Senior Technology Acceptance Model (STAM) and the Non-adoption, Abandonment, Scale-up, Spread, and Sustainability (NASSS) framework.

3.3.1 Research Stages

As described in Figure 5, the following stages will be followed throughout the study:

1. A heuristic evaluation of the existing Next-of-Kin Compass prototype to identify baseline usability issues and those relevant to older users. This supports RQ1 by identifying usability and ageing-related barriers.

2. A reflection workshop with caregiving and design researchers to contextualize the previous usability findings within caregiving in Sweden and designing for the elderly. This supports RQ1 by interpreting ageing and usability barriers in context.

3. The redesign of the Next-of-kin Compass to tackle usability findings from previous stages. This supports RQ2 and RQ3 by translating barriers to concrete design choices.

4. Semi-structured interviews with multiple stakeholder groups to explore the broader caregiving context, especially as an older person, and reflect upon the relevance and appropriateness of key new features in the prototype redesign. This supports RQ1 and RQ2 by analyzing challenges and design choices through stakeholder perspectives.

5. Data analysis using STAM and NASSS as interpretative lenses. This supports RQ1 by outlining key ageing and caregiving challenges through individual and systemic perspectives.

6. A synthesis of design guidelines, which supports RQ2 by translating challenges to design choices.

7. A second iteration of the prototype redesign to operationalize the design guidelines, which supports RQ3.

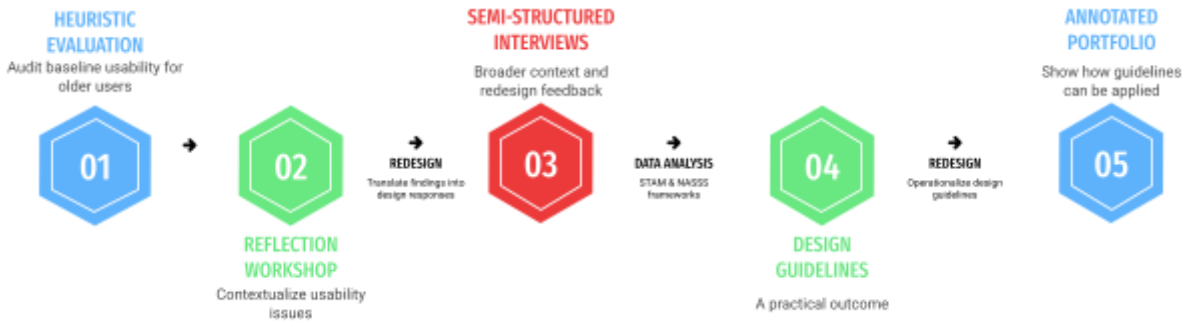


Figure 5. Research Process. This figure shows the process the research follows from left to right. It explains what is done at each step and its purpose.

The subsequent subsection provides a detailed account of each data collection method employed in the study.

3.4 Data Collection

The following sub-sections detail each of the methods used in the study: a heuristic evaluation, an expert reflection workshop, semi-structured stakeholder interviews, and an iterative prototype redesign. For each method, it is outlined how they were executed, who was involved, and their specific analytical purpose within the broader study.

3.4.1 Heuristic Evaluation

Heuristic evaluation is a method in which usability experts systematically assess a digital tool against a set of design principles or heuristics to identify specific usability issues (Nielsen, 1994). In this study, the evaluation was conducted independently by the study authors, given their experience and expertise in software development and usability evaluation.

It is common to rely on established guidelines such as Nielsen's (1994) ten usability heuristics, but custom heuristics can be created to assess usability in specific cases. In this case, custom heuristics were created by the study authors to identify usability issues for older users within the Next-of-Kin Compass prototype. To create these custom heuristics, the study authors conducted a background review of ageing-related design guidelines and recommendations. Then, the most prevalent considerations were mapped against Nielsen's usability principles, as can be observed in the following table.

Table 1. Custom Heuristics for Usability Among Older Adults.

Heuristic	**Description**
1. Visibility of System Status	The system should always keep users informed about what is happening through appropriate and timely feedback. For older adults, immediate and clear status changes minimize uncertainty and prevent anxiety, compensating for age-related declines in divided attention and working memory (Liu et al., 2021; Chirayus & Nanthaamornphong, 2019).
2. Match Between System & Real World	The interface must be aligned with the user's language. This means using familiar, simple, and clear words, phrases, and concepts rather than technical jargon. For older adults, this reduces the cognitive burden and improves comprehension (Liu et al., 2021; Morey et al., 2019; Gomez-Hernandez et al., 2023).
3. User Control & Freedom	The system should provide exits for when users

	trigger functions by mistake. For older users, who may experience age-related motor declines or tremors, persistent return options and flat navigation tracks mitigate the frustration of accidental actions (Liu et al., 2021; Gomez-Hernandez et al., 2023).
4. Consistency & Standards	The system should be generally consistent, so users do not wonder whether different words, situations, or actions mean the same thing. Maintaining uniformity across visual cues and terms allows older adults to build reliable mental models, lowering the cognitive load needed to navigate interfaces (Liu et al., 2021; Tajudeen et al., 2022).
5. Error Prevention	The system should prevent problems from occurring. Designing systems that are mindful of possible errors is essential for older adults to accommodate motor-related interaction errors such as slips, precision tracking issues, or accidental gestures (Amouzadeh et al., 2025; Wildenbos et al., 2018).
6. Recognition Rather than Recall	The system should minimize the user's memory load by making elements, actions, and options highly visible. Older adults benefit significantly from visible main functions and flat hierarchies, eliminating the need to hold previous multi-step navigation paths in their short-term memory (Liu et al., 2021; Morey et al., 2019).
7. Flexibility & Efficiency of Use	To support ageing populations with diverse physical and sensory capabilities, flexibility means offering adjustable interfaces and alternative modalities for interaction like audio or visual backups (Liu et al., 2021; Amouzadeh et al., 2025; Harshini Ramdowar et al., 2023; McCarthy et al., 2024; Khamaj, 2024).
8. Aesthetic & Minimalist Design	Interfaces should not contain information that is irrelevant or rarely needed. For older adults experiencing natural age-related changes in vision and visual processing, uncluttered layouts with large text and high contrast prevent cognitive overload and improve text legibility (Liu et al., 2021; Morey et al., 2019).
9. Help Users Recover from Errors	Error messages should be expressed in plain language, precisely indicate the problem, and constructively suggest a solution. Clear and concise recovery prompts written in plain

	language ensure older adults can comprehend what went wrong and resolve issues without feeling lost (Gomez-Hernandez et al., 2023; Wildenbos et al., 2018).
10. Help & Documentation	The system should provide help that is easy to search and focused on the user's task. Older adults generally favor structured and explicit guidance such as step-by-step instructions or video tutorials over trial-and-error exploration (Gomez-Hernandez et al., 2023; Wildenbos et al., 2018).

Having these guidelines, the study authors individually interacted with the prototype, exploring different use cases. Then, they documented usability violations across each criterion and consolidated their findings into a shared list.

The heuristic evaluation allowed us to identify potential usability challenges for older users within the existing prototype. The findings generated a baseline interpretation of the prototype regarding interface-level challenges for older users. Additionally, the findings informed subsequent workshop discussions and redesign priorities.

3.4.2 Reflection Workshop

Following the heuristic evaluation, a half-day reflection workshop was led by the study authors with two participants. Reflection workshops facilitate collaborative interpretation and contextualisation of findings through discussion among participants with relevant expertise (Jenkins et al., 2016). In this study, the workshop was used to examine the relevance, implications and possible solutions for identified usability issues in relation to broader caregiving realities, ageing-related challenges, the Swedish healthcare context, and adoption drivers among caregivers.

The first participant was a researcher in caregiving (CR1), who specializes in informal caregiving and family support in the Swedish context. The second was a researcher in design (DR1), with experience designing for caregiver support. Including participants with these different expertise domains strengthened the contextual interpretation of the findings and reduced reliance on a single disciplinary perspective.

During the workshop, participants reviewed the heuristic evaluation findings. They discussed the relevance of specific usability issues for caregivers in Sweden. In addition, participants reflected on potential design responses to the identified issues so that they are appropriate for older caregivers in Sweden. Notably, the discussions enabled the identification of broader socio-technical factors influencing digital health adoption among older caregivers in Sweden.

Insights generated through the workshop informed both the redesign of the prototype and the development of topics explored during the subsequent stakeholder interviews.

3.4.3 Prototype Redesign

Prototype redesign functioned as both a design activity and a research activity throughout the study. Following RtD principles, redesign was not treated solely as an

outcome but as a mechanism for exploring, materialising, and refining emerging knowledge. Redesign also enabled the study to investigate not only what challenges existed but also how potential design responses could be practically implemented.

Redesign by the study authors was conducted in two stages of the study, as observed in Figure 5. First, the prototype was changed after the reflection workshop to tackle the most relevant design issues found. The outcome of this redesign served as a probe for exploring and reflecting upon the appropriateness of new features with stakeholders during the interviews. The second redesign happened after the design guidelines were outlined in order to materialize them, showing their applicability in a real-world application. Thus, the movement between analysis and redesign strengthened coherence between empirical findings and resulting design interventions while supporting the generation of context-sensitive and practice-oriented design knowledge.

3.4.4 Semi-Structured Interviews

Semi-structured interviewing is a method that enables participants to describe their experiences, perspectives, and interpretations while allowing researchers to flexibly explore emerging topics and contextual issues during the interview process (Kallio et al., 2016). This flexibility was particularly important given the exploratory nature of the study and the complexity of caregiving and ageing experiences. This method allowed for a broader understanding of ageing and caregiving challenges in the Swedish context and the appropriateness of key design responses in tools such as the Next-of-kin Compass.

The interview guides, as observed in Appendix 9.1, were guided by the study's research question key topics such as caregiving and ageing challenges, but also were informed by previous findings across the heuristic evaluation and the reflection workshop. In addition, the interview guides used the redesigned Next-of-kin prototype as a probe for reflecting upon key design features to tackle usability issues found in previous stages.

The interviews were conducted with participants from multiple stakeholder groups, including:

- Two researchers in caregiving. The first had a background in ageing and family support (CR2). The second had experience in cancer caregiving (CR3).
- A researcher in Human-Computer Interaction with expertise in interaction design and evaluation (DR2).
- A designer with experience in digital health (DR3).
- A representative from the National Competence Centre for Carers in Sweden (NKA) who is also a researcher in healthcare technology for the elderly and their relatives.
- A representative from Cancerkompisar, a non-profit organization that supports caregivers of cancer patients in Sweden (CC).
- An older adult with lived experience caring for a family member with cancer (CG1).

The inclusion of multiple stakeholder groups enabled the study to capture complementary perspectives regarding broader caregiving, ageing-related challenges, system-level factors, and digital health design. This strengthened the credibility and analytical depth of the findings by allowing patterns to be examined across different forms of expertise and experience.

Participants were recruited using purposive sampling and professional networks to ensure the inclusion of individuals with relevant experience related to caregiving, ageing, digital health, or interaction design. Purposive sampling was considered appropriate because the study prioritised information-rich participants capable of contributing detailed and contextually relevant insights rather than statistically representative sampling.

The inclusion criteria differed across participant groups:

- Older caregivers were required to be aged 65 years or older and have experience providing unpaid care to a family member with cancer.
- Design and digital health participants were required to have professional or research experience in HCI, UX design, interaction design, or digital health.
- Caregiving stakeholders were required to have professional or research experience related to caregiving, ageing, support services, or cancer care.

The study initially aimed to recruit approximately 8–12 participants across stakeholder groups, a range considered appropriate for a qualitative study focused on depth and iterative inquiry (Malterud et al., 2016).

The final sample consisted of eight participants representing caregiving research, HCI and digital health expertise, caregiver support organisations, and lived caregiving experience. Although only one older caregiver participant was ultimately recruited, the study balanced this limitation through the inclusion of multiple complementary stakeholder perspectives and by prioritising information-rich participants. In addition, recruiting representatives from caregiving organizations allowed understanding perspectives beyond individual experiences, which come from interacting with many caregivers over a long period of time.

Interviews lasted approximately 30–45 minutes and were conducted in person and through video conferencing according to participant preference. All interviews were audio-recorded with informed consent and later transcribed for analysis.

3.5 Data Analysis

All qualitative data collected throughout the study were analysed using thematic analysis following the six-phase framework proposed by Braun and Clarke (2006). The analysis was conducted iteratively across the different stages rather than as a single post-hoc activity. This enabled emerging insights to continuously inform subsequent design decisions, prototype revisions, and reflections throughout the study.

The analysis process began with repeated familiarisation with the empirical material, including interview transcripts, workshop reflections, field notes, heuristic evaluation observations, and redesign documentation. Initial coding was then conducted systematically across the dataset to identify recurring usability concerns, caregiving

and ageing challenges, and contextual factors influencing engagement with digital tools.

This stage combined semantic coding, capturing explicit statements and observable issues, with latent coding to interpret underlying assumptions, tensions, and behavioural patterns related to caregiving and digital health use. Combining semantic and latent analysis strengthened the analytical depth of the study by enabling both visible usability problems and broader socio-technical dynamics to be examined simultaneously.

Examples of semantic codes included small font size, limited time for app use, preference for paper-based tracking, and physical tiredness. Latent coding captured broader interpretive patterns such as cognitive overload, caregiving responsibility anxiety, dependence on memory, and reduced confidence in interacting with digital systems.

Codes were continuously compared, refined, and reorganised throughout the analytical process. The codes were subsequently grouped into broader candidate themes representing recurring patterns across data sources. These themes were iteratively reviewed and refined in relation to both the empirical material and the evolving redesign process.

Rather than treating themes as fixed categories, the analysis focused on relationships between ageing-related constraints, caregiving responsibilities, emotional burden, healthcare system complexity, and digital health engagement. This supported the generation of design-oriented knowledge by identifying how usability challenges emerged within the lived realities of older caregivers.

To strengthen conceptual coherence, the themes were further interpreted through the Senior Technology Acceptance Model (STAM) and the Non-adoption, Abandonment, Scale-up, Spread, and Sustainability (NASSS) framework. These frameworks were not used as predefined coding structures but as interpretive lenses during later analytical stages to contextualise the findings within broader discussions of ageing, caregiving, and socio-technical digital health complexity.

3.5.1 The NASSS Framework: System-Level Complexity in Caregiving Contexts

To analyze how system-level complexity shapes digital health use in caregiving contexts, this study applied the Non-adoption, Abandonment, Scale-up, Spread, and Sustainability (NASSS) framework developed by Greenhalgh et al. (2017). The framework conceptualizes adoption as the outcome of interacting technological, organizational, and contextual factors.

This perspective is particularly relevant in digital health, where technologies are often introduced with implicit assumptions that innovation will automatically translate into improved care efficiency and outcomes (Greenhalgh et al., 2017). NASSS challenges this assumption by showing that adoption and sustained use are contingent on the interaction between multiple layers of complexity across users, technologies, organizations, and broader institutional environments.

In this study, NASSS was used as an analytical framework for examining how caregiving contexts shape digital health adoption across multiple interconnected domains. Rather than limiting analysis to a subset of domains, this study engaged with

the full NASSS structure in a flexible and iterative manner, with particular attention to the following domains:

- **The Condition:** the complexity, instability, and lived experience of caregiving, including emotional strain and fluctuating care demands
- **The Technology:** the material and interactional complexity of digital tools, including usability, adaptability, and integration requirements
- **The Value Proposition:** the perceived and institutional value of digital tools from both caregiver and healthcare system perspectives
- **The Adopter System:** the identities, capacities, and roles of caregivers and healthcare professionals, including informal and distributed responsibility
- **The Organisation:** the capacity of healthcare systems to integrate and sustain digital tools within existing workflows and infrastructures
- **The Wider Context:** policy, regulatory, and socio-institutional factors shaping implementation and scalability

Across these domains, NASSS was used not as a rigid classification system but as a way of tracing how complexity emerges through interactions between caregiving realities and healthcare system structures. This allowed the analysis to capture not only isolated barriers but also how different layers of the system reinforce or amplify each other in shaping digital health engagement.

3.5.2 Senior Technology Acceptance Model (STAM): Individual Capacity Constraints

The Senior Technology Acceptance Model (STAM), as observed in Figure 6, was used in this study to examine how age-related changes shape older adults' capacity to engage with digital health technologies at the individual level. Unlike traditional technology acceptance models, STAM explicitly incorporates age-related factors such as health status, cognitive self-efficacy, gerontechnology anxiety, and functional ability in explaining technology use. This is particularly relevant in caregiving contexts, where older caregivers must engage with digital systems while simultaneously managing their own age-related limitations and the demands of providing care. Therefore, within this study, STAM provided the analytical basis for examining how aging affects the baseline capacity available for the engagement of older caregivers.

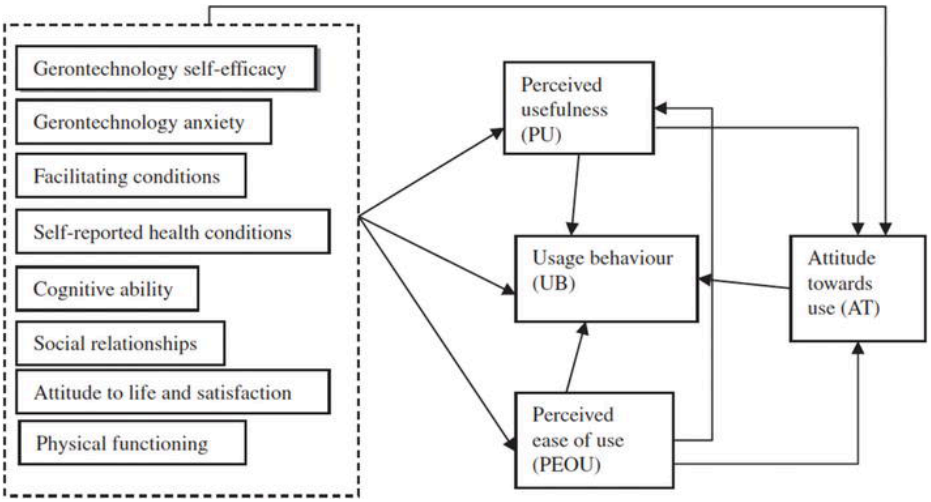


Figure 6. Senior Technology Acceptance Model (STAM). This figure shows the age-related attributes that impact technology acceptance for older users. *Note*: Adapted from "Use of Senior Technology Acceptance Model (STAM) for Social Robots Studies," by K. Cheng, 2024, Current State and Future Perspective in Human-Robot Interaction, IntechOpen (https://doi.org/10.5772/intechopen.1007144). CC BY 3.0.

3.6 Ethical Considerations

Following consultation with the academic supervisor and current institutional guidelines, formal approval from the Swedish Ethical Review Authority was not required for this study. Nevertheless, an internal structured ethical assessment was conducted prior to data collection to ensure compliance with institutional research ethics standards and to align the thesis procedures with the ethical approval framework used within the broader FamCASP research initiative.

Several ethical considerations were particularly relevant given the study's focus on older caregivers and caregiving-related experiences. Potential risks included emotional discomfort associated with discussing caregiving experiences, confidentiality concerns, and accessibility challenges related to digital participation.

To mitigate these risks, participants received detailed information regarding the study's purpose, procedures, voluntary participation, and data handling practices before participation. Informed consent was obtained from all participants. Participants were additionally informed that they could pause or withdraw from the study at any time without consequence.

All procedures complied with GDPR and the Swedish Research Council's guidelines for good research practice. Personal identifiers were stored separately from research data, transcripts were anonymised during transcription, and digital data were stored on encrypted drives accessible only to the research team.

Reflexivity and researcher positionality additionally formed part of the ethical and methodological considerations of the study. Throughout the research process, the researchers maintained reflexive journals documenting assumptions, emotional responses, analytical interpretations, and methodological decisions. Reflexive practice was employed not to eliminate subjectivity but to acknowledge the researchers' active

role within the interpretive process and strengthen the transparency and confirmability of the findings (Noble & Smith, 2025).

The researchers' backgrounds in design, digital health, and person-centred care inevitably shaped the analytical process and interpretation of findings. Maintaining ongoing reflexive awareness and detailed documentation of analytical decisions strengthened the traceability and trustworthiness of the research process.

Although the interpretive nature of qualitative analysis means that findings are partially shaped by researcher interpretation, transparency, triangulation, documentation, and reflexive practices were employed throughout the study to strengthen methodological dependability and confirmability.

3.7 Alternative Methods

Several alternative methodological approaches were considered during the design of the study. First, a purely qualitative approach based exclusively on ethnography or semi-structured interviews could have generated rich descriptions of caregiving experiences and digital health practices. However, such approaches are primarily descriptive and offer limited support for systematically translating findings into concrete and iteratively refined design interventions (O'Brien et al., 2016). Given that the study aimed not only to understand challenges but also to explore design responses, a more generative methodological approach was required.

A conventional user-centred design (UCD) process was additionally considered because of its emphasis on iterative usability refinement. However, traditional UCD approaches often prioritise product optimisation over the generation of transferable design knowledge and may focus narrowly on interface-level interactions while underexamining broader socio-technical contexts (van Velsen et al., 2022). Since the present study aimed to investigate how caregiving and ageing collectively shape broader digital health design considerations, a wider analytical lens was necessary.

Participatory and co-design approaches were also considered because they can strengthen democratic participation and ensure closer alignment with users' lived experiences (Rothausen et al., 2025). Nevertheless, sustained participation among older caregivers of cancer patients presents practical and ethical challenges, including emotional burden, fluctuating health conditions, limited availability, and recruitment difficulties. Previous research additionally highlights that extensive participation demands may become unrealistic for vulnerable populations with age-related or caregiving-related limitations (Cornet et al., 2020). Consequently, the study adopted a more pragmatic approach that balanced participant involvement with participant capacity. Stakeholder perspectives were incorporated through interviews and reflective activities while avoiding intensive long-term participation requirements that could impose excessive burden on participants.

An experimental usability-testing approach was also considered. Controlled usability testing could have generated quantitative measures related to task completion, efficiency, or interface performance. However, the present study prioritised exploratory understanding of caregiving and ageing-related challenges over quantitative performance measurement. A generative and interpretive methodology was therefore considered more appropriate than a primarily evaluative experimental design.

Finally, a multiple-case study design could have enabled comparison across several caregiving tools or healthcare contexts. However, the present study prioritised depth and contextual understanding over comparative breadth by focusing on a single embedded case within its real-world development environment. This allowed for more detailed investigation of the socio-technical interactions shaping digital health engagement among older caregivers.

4. Results

This section presents the findings from the four stages of the study and demonstrates how each stage informed the next. First, a heuristic evaluation of the Next-of-Kin Compass prototype was conducted to identify usability issues affecting older caregivers. Second, these findings were reviewed and discussed during a reflection workshop with a design researcher and a caregiver researcher. Third, insights from both activities informed a redesign of the prototype. Finally, semi-structured interviews were conducted with caregiving stakeholders, design experts, and one caregiver with lived experience to get feedback on the proposed design improvements and explore broader factors influencing the adoption of digital caregiving tools.

4.1 Heuristic Evaluation

The heuristic evaluation identified 19 usability issues across Nielsen's ten heuristics adapted for older adults. The full list of issues is provided in Appendix 9.2, Table 2. Issues were initially documented against Nielsen's ten heuristic principles and subsequently grouped into five thematic categories to facilitate understanding: cognitive burden, lack of guidance and feedback, limited user control, accessibility limitations, and navigation issues.

- **Cognitive Burden -** The evaluation identified interface elements with unclear or uninformative labels. Generic labels such as "Read More" on the results screen did not communicate what content would appear after selection. Assessment scale options such as "Moderately" and "Seriously" concerned were not clearly differentiated, making it difficult to distinguish between adjacent response options. The "Column" and "Chart" view labels on the results screen provided no indication of what each visualisation would display.

- **Lack of Guidance and Feedback -** Some areas of the prototype provided limited guidance or feedback during evaluation. No onboarding support was available for first-time users, meaning the purpose and structure of the tool were not explained on first access. Following account registration, no message confirmed that a verification email had been sent. No confirmation message appeared following the logout. Password creation instructions did not explain what "special characters" referred to or provide examples. When a screening was in progress, no explicit save, cancel, or reset buttons were present, making it unclear whether progress was being retained.

- **Limited User Control -** The evaluation identified limitations in the degree of control available when managing interactions with the system. No explicit options to save or cancel a screening in progress were visible, despite the screening consisting of multiple steps. To change a previously submitted response, the evaluators were required to navigate through every step of the

screening again rather than editing the relevant response directly. Reminder-related functions did not allow the specific time of email reminders to be viewed or adjusted, meaning users could not control when notifications would be received.

- **Accessibility Limitations -** The prototype presented accessibility-related limitations across multiple areas. Text-size adjustment was not available anywhere within the interface. No audio playback was available for informational articles or recommendation content. Note-taking relied solely on manual text entry, with no voice input option available. Recommended external sources linked from the results screen were general in nature, for example linking to broad informational pages rather than content specific to the user's screening results. Some sections, particularly the screening and results pages, required substantial vertical scrolling to access all content.

- **Navigation Issues -** The evaluation identified issues concerning the discoverability of features and navigation throughout the system. On mobile devices, key functions including notes and reminders were not immediately visible on the homepage and required additional steps to locate. Certain elements styled as bold text were not distinguishable as interactive, while some elements styled as buttons or cards appeared clickable but were non-interactive. The application used inconsistent terminology, referring to the same feature as both "Survey" and "Screening" in different places across the interface. When a new screening was started, the most recent result disappeared from the home screen, removing access to previously available recommendations and graph views.

The findings from the 19 usability issues were subsequently presented during the reflection workshop, where participants reviewed the identified issues and explored potential design responses.

4.2 Reflection Workshop

As described in the Methodology (Section 3.4.2), the workshop included a caregiving researcher (CR1) and a design researcher (DR1). The workshop was conducted as a semi-structured discussion. Participants were first presented with the usability issues identified during the heuristic evaluation and were asked to reflect on their relevance, severity, and possible solutions. The discussion then broadened to consider factors beyond interface design that may affect whether caregivers adopt and continue using digital support tools. Notes from the workshop were organised into two overarching areas: (1) prioritisation of usability issues and (2) broader contextual factors influencing adoption.

4.2.1 Prioritisation of Usability Issues

Both CR1 and DR1 largely agreed with the usability issues identified during the heuristic evaluation and provided additional context regarding their causes. A recurring theme concerned the clarity of information and navigation. CR1 and DR1 acknowledged that several interface elements relied on assumptions about users' familiarity with digital systems. For example, labels such as "Read More" were considered insufficiently descriptive, requiring users to infer what information would appear after selection. Similarly, the purpose of different graph views was considered unclear and both participants agreed additional explanation would be beneficial.

The workshop also highlighted issues related to user control and feedback. Both participants agreed that clearer confirmation was needed for actions such as account registration, reminder scheduling, and saving screening progress. CR1 noted that although some functionality existed in the background, users could not easily determine whether their actions had been successfully completed.

Several observations focused on recognition and accessibility. CR1 and DR1 noted that some interface elements relied solely on visual styling such as bold text to indicate functionality, creating ambiguity regarding which elements were interactive. Non-interactive elements styled as buttons or cards were also identified as potentially misleading.

The participants also identified opportunities to increase flexibility and accessibility. Suggestions included providing audio alternatives for longer text sections, offering greater personalisation of recommendations, and allowing users to customise aspects of the interface. While agreeing that some of these features would require additional technical development, both participants considered them valuable directions for improvement.

Even so, not all identified issues were viewed as requiring immediate redesign. CR1 noted that extensive scrolling, while acknowledged as a concern, could not be straightforwardly resolved, as reducing scrolling would likely increase navigation complexity through additional clicks and page transitions.

Overall, the workshop confirmed many of the heuristic findings while helping distinguish between issues that required immediate redesign and those that reflected broader design trade-offs.

4.2.2 Contextual Factors Influencing Adoption

Beyond interface usability, CR1 and DR1 identified broader contextual factors that may influence whether caregivers adopt and continue using a digital support tool. Psychological readiness was identified as one factor. Participants noted that caregivers often experience high levels of stress, emotional burden, and uncertainty and that these conditions may affect engagement with digital tools during particularly demanding periods. In addition, digital literacy was identified as a relevant factor. CR1 and DR1 noted that older caregivers vary considerably in their familiarity with digital technologies and that limited digital confidence, language barriers, and previous negative experiences with technology may create obstacles to adoption.

Trust and data privacy also emerged as relevant considerations. Both participants noted that transparent communication regarding how personal information is stored and used was considered important for building user trust. Similarly, CR1 and DR1 highlighted the importance of professional endorsement for fostering trust in users. CR1 and DR1 noted that tools introduced or recommended by healthcare professionals were more likely to be adopted than those discovered independently.

Findings from the reflection workshop informed the redesign of the Next-of-Kin Compass prototype. Accordingly, the following section describes how the prioritised issues were translated into specific design modifications.

4.3 Prototype Redesign

Rather than attempting to address every identified usability issue, the redesign focused on those considered most likely to improve the experience of older caregivers while remaining feasible within the scope of the study.

Thus, the selected redesign priorities reflected three overarching goals: improving clarity and predictability, reducing cognitive and physical effort, and increasing accessibility for users with diverse needs and levels of digital experience. The following subsections describe each redesign decision, the evidence supporting it, and its intended benefit for older caregivers.

- **Descriptive Navigation Labels -** The heuristic evaluation identified that "Read More" labels throughout the results interface did not communicate what content would appear after selection. During the reflection workshop, CR1 acknowledged that the existing label could be improved to better function as a recommendation or prompt toward help, and DR1 confirmed that the lack of clarity was relevant to the tool's refinement. On this basis, the labels were replaced with descriptive alternatives such as "See Recommended Advice," making the purpose of each link explicit before selection (Figures 7–8).

- **Personalised Content Based on Screening Results -** The heuristic evaluation noted that recommended resources were not tailored to individual screening responses, meaning users received the same information regardless of their specific situation. During the reflection workshop, CR1 acknowledged that more targeted resources would improve relevance, while also proposing a layered approach to ensure that users with needs outside defined categories still had access to general information. Personalised recommendations derived from each user's screening results were therefore introduced, surfacing the most relevant resources at the point of need (Figures 9–10).

- **Audio Playback for Informational Content -** The heuristic evaluation identified that the prototype offered no alternative to reading lengthy text sections, with no audio playback available for informational articles or recommendation content. During the reflection workshop, DR1 noted that presenting content in accessible formats reduces the risk of overwhelming users. On this basis, audio playback was added to recommendation articles and longer informational content (Figures 11–12).

- **Voice-Note Input for Personal Reflections -** The heuristic evaluation identified that the notes feature relied solely on keyboard text entry. During the reflection workshop, CR1 acknowledged that the notes function had been introduced late in the design process but that it could serve as a central feature for many users, particularly those using the tool on a phone. Voice-note input was therefore introduced as an alternative method for capturing personal reflections (Figures 13–14).

- **Onboarding Guidance -** The heuristic evaluation identified the absence of any structured introduction to the prototype's features, noting that users were expected to discover functionality independently. During the reflection workshop, CR1 acknowledged that the current approach was minimal and confirmed that a walkthrough of features would be feasible and beneficial. A step-by-step onboarding flow was therefore introduced to guide users through

the main features when they first access the prototype (Figure 15), with the design kept deliberately brief.

In summary, the redesign addressed five prioritised usability issues identified through the heuristic evaluation and reviewed during the reflection workshop. Collectively, the changes aimed to reduce cognitive burden, improve accessibility, and lower the effort threshold for older caregivers interacting with the tool. Generic labels were replaced with descriptive alternatives, recommendations were personalised to screening results, audio playback and voice-note input were introduced as multimodal alternatives to text-based interaction, and a brief onboarding flow was added to support first-time users. Together these modifications represented a targeted response to the most impactful barriers identified in the earlier phases of the study. The redesigned prototype subsequently served as a discussion artefact during the semi-structured interviews, enabling participants to reflect on the appropriateness and potential usefulness of the proposed changes.

4.4 Findings from Semi-Structured Interviews

Following the redesign phase, semi-structured interviews were conducted to explore how the proposed design improvements aligned with the needs, experiences, and expectations of caregivers and caregiving professionals. The redesigned prototype served as a discussion artefact during the interviews, enabling participants to reflect on the usefulness of the proposed features while also discussing broader challenges associated with caregiving and digital support tools.

The interview data were analysed using Braun and Clarke's (2006) thematic analysis approach. Following familiarisation with the data, transcripts were coded and organised into recurring patterns and themes. Four themes were identified:

1. The caregiving context
2. User pain points in digital caregiving tools
3. Feedback on proposed design improvements
4. Factors influencing adoption

The following sections present these themes and illustrate how participants' experiences and reflections contribute to understanding the design of digital support tools for older caregivers.

4.4.1 The Caregiving Context

This theme captures emotional strain, practical coordination demands, and the challenges caregivers face when navigating fragmented support systems.

Participants characterised caregiving in cancer contexts as a complex role involving emotional, practical, and coordination responsibilities. Caregivers often act as central coordinators of the patient's daily life, managing appointments, medications, and communication with healthcare providers. As one caregiving researcher (CR2) explained:

> *"Relatives take on practical, emotional, and coordination tasks… driving them to the hospital, managing side effects, and being in contact with healthcare."* (CR2)

In addition to these responsibilities, participants highlighted that caregivers often experience significant strain. Another caregiving researcher (CR1) noted that “many caregivers are overwhelmed” (CR1). The lived-experience caregiver (CG1) similarly described caregiving as a continuous, long-term responsibility:

> *“I took care of her for four and a half years… twenty-four seven.”* (CG1)

Participants also described the emotional impact of a cancer diagnosis on families. The caregiver reflected on the initial shock of receiving the diagnosis:

> *“You cannot take in the message… it was very brutal.”* (CG1)

Researchers further noted that cancer is often associated with uncertainty and existential concerns, even when the prognosis is positive:

> *“Cancer is so closely related to death… even when the prognosis is good, families start thinking about the future alone.”* (CR1)

In addition to emotional strain, older caregivers were frequently described as particularly vulnerable. Participants reported that many older caregivers face their own health challenges alongside caregiving responsibilities, including sensory and cognitive limitations.

> *“Many older caregivers may be ill themselves… and they have physical limitations, vision problems, hearing problems, or cognitive issues that make everything harder.”* (CR2)

Participants also noted that older caregivers often manage caregiving independently and may avoid seeking support from family members.

> *“Older people often deal with it by themselves. They don’t want to disturb their children.”* (CR2)

Another challenge identified was that many individuals do not recognise themselves as caregivers within the healthcare system. Instead, they often search for information based on the patient’s condition rather than their own support needs. The caregiver support organisation director (CC) noted:

> *“They don’t know they are ‘next of kin’… they search for ‘my husband has cancer,’ not ‘I need support.’”* (CC)

Beyond these individual experiences, participants described the support system as fragmented and difficult to navigate. Caregivers often need to interact with multiple services across healthcare and municipal systems. The representative from the Swedish National Competence Centre for Relatives (NKA) noted:

> *“In Sweden currently it is a jungle… many carers describe it as a challenge to navigate the system.”* (NKA)

Participants also noted that early hospital discharge increases the responsibility placed on family members, as recovery continues at home. A caregiving researcher (CR3) stated:

> *"Patients are discharged very early… recovery continues at home, and family members become informal caregivers."* (CR3)

The caregiver participant described feeling emotionally unsupported in this transition:

> *"Nobody asked me how I feel… it was just a routine question."* (CG1)

Generational attitudes were also identified as influencing help-seeking behaviour. Some participants noted that older individuals may avoid contacting healthcare services to prevent burdening others:

> *"Older persons hesitate to contact healthcare because they don't want to cause trouble."* (CR3)

As a result, caregivers may delay seeking help for themselves while prioritising the patient's needs:

> *"They put their needs on pause… and then suddenly they don't know where to turn."* (CR1)

Overall, participants described caregiving as a demanding role involving emotional strain, practical responsibilities, and challenges in navigating fragmented support systems.

4.4.2 User Pain Points in Digital Caregiving Tools

This theme outlines the challenges older caregivers encounter when interacting with digital tools. It focuses on barriers related to complexity, cognitive load, accessibility, emotional exhaustion, and mismatches between system design and user expectations.

Participants repeatedly emphasized that complexity and high cognitive load act as major barriers for caregivers who are already overwhelmed by their daily responsibilities. As CR1 noted regarding app navigation:

> *"If it's ten clicks before you get to the information, it's not working."* (CR1)

This friction is intensified by the emotional and cognitive exhaustion many older caregivers experience. Experts explained that this group often has limited mental capacity left for navigating digital systems, making even simple tasks feel burdensome.

Memory retention issues further complicate engagement for older caregivers. As CR2 described:

> *"Even if you explain how to use it, next time they may not remember… then it's not a help."* (CR2)

Participants also highlighted sensory and motor limitations that can hinder interaction with digital tools:

> *"Vision can be a problem, hearing can be a problem… and cognitive function can be a problem, which means they don't understand simple things."* (CR2)

Other interviews reinforced these barriers. The caregiver support organization director emphasized that older adults often struggle with basic smartphone interactions:

> *"The tech around smartphones is always tricky… they need help to find things."* (CC)

Even accessing digital services could present barriers. For example, QR codes were described as difficult to use:

> *"A QR code is a great obstacle… I wouldn't use it myself."* (CR3)

The caregiver participant also described difficulties assessing the reliability of online information:

> *"The internet is full of misleading messages and advice… how do you know someone is serious?"* (CG1)

Several experts also noted that scrolling can be difficult for older adults, especially when long pages require sustained attention or precise motor control. CR3 explained that excessive scrolling may overwhelm users who already feel cognitively taxed.

Another pain point was the mismatch between designers' assumptions and users' mental models. CR2 explained that interfaces that appear logical to designers may not be intuitive for older adults:

> *"It's probably logical for the people who designed it, but not for me."* (CR2)

The caregiver support organization director added that unfamiliar UI patterns can further increase confusion, as older adults rely heavily on predictable, familiar interaction styles.

The HCI researcher (DR2) added that the emotional weight of the screening questions themselves can become a usability barrier:

> *"Filling out the questions can make you realize how much you're dealing with… it can be emotionally heavy."* (DR2)

DR2 also noted that fragmentation across digital systems contributes to usability challenges. When tools sit outside familiar platforms such as 1177, caregivers must remember new URLs, new access points, and new interaction patterns all of which increase cognitive load.

Participants further emphasized the risk of overwhelming caregivers with too much information presented at once. As digital health design researcher (DR1) noted:

> *"Don't bombard people with information… blank space is important so it doesn't become overwhelming."* (DR1)

They also described situations where digital tools may increase rather than reduce caregiver burden:

> *"The technology has not improved the situation… it has been an added challenge for the person being cared for and an added challenge for the*

informal caregiver." (NKA)

At the same time, participants highlighted variation in digital literacy among older caregivers. The caregiver participant described frequent digital engagement:

> *"I use the Internet frequently… information search, entertainment, interactive museums, travel experiences."* (CG1)

He also described using digital tools for documentation and archiving:

> *"I write a diary every single day… I upload it to a USB stick and keep it in a safe."* (CG1)

Together, these insights highlight the diverse usability barriers older caregivers encounter when engaging with digital tools, highlighting how cognitive, sensory, emotional, and technical factors shape their digital experiences.

4.4.3 Feedback on Proposed Design Improvements

This theme summarises expert reflections on information presentation, personalisation, accessibility features, and onboarding strategies within the redesigned tool.

One feature discussed was the use of "Read More" labels that allow users to expand information progressively. Although the redesign proposed changing the label to a more explicit phrase such as "See Advice," and one design researcher agreed this wording could make the action clearer for older adults, another expert shifted the discussion toward the broader cognitive value of the mechanism itself. They emphasized that the key benefit lies in reducing information overload by presenting content in smaller, manageable sections, rather than in the specific phrasing of the label. As one expert explained:

> "*Breaking information into smaller sections helps users focus on what they need without feeling overwhelmed.*" (DR1)

Personalisation was generally seen as beneficial when used to prioritise relevant information:

> "*If the system highlights the information that is most relevant, it saves time.*" (CR1)

However, participants also highlighted challenges associated with collecting sufficient data for personalisation:

> "*It's difficult to make something personalized without asking for a lot of personal information… that's a dilemma.*" (CR1)

Onboarding was emphasised as a critical design moment and should remain simple and brief:

> "*Don't ask too many questions upfront… they just want to sign up.*" (DR2)

Participants also responded positively to the idea of providing audio alternatives for

longer texts, suggesting that this could support users who experience reading fatigue or visual difficulties:

> "*Audio options can make information more accessible, especially for users who may find long texts difficult to read.*" (DR1)

Both the caregiver support organization director and surgical care researcher strongly supported audio and voice-based input:

> "*Voice notes make the tool more inclusive.*" (CR2)

> "*Audio would help tired users who don't have the energy to read.*" (CR3)

In addition to accessibility benefits, interviews also highlighted that multimodal options are valuable even for users without impairments, as some caregivers simply prefer listening over reading or find audio more convenient in emotionally or physically demanding situations.

Experts also emphasized the importance of onboarding processes that guide users through the functionality of the tool when they first begin using it. However, the surgical care researcher stressed that onboarding must be extremely short:

> "*The introduction must be very, very short… it cannot be an obstacle.*" (CR3)

Regarding the screening tool, the HCI researcher highlighted the importance of sequencing questions to reduce emotional burden:

> "*Don't start with the hardest questions… it increases emotional load.*" (DR2)

Beyond the screening flow, the HCI researcher noted that caregivers prefer simple, immediate representations of their needs rather than complex data visualizations or long-term trend graphs. They emphasized that caregivers want to focus on the "here and now," using clear, color-coded categories rather than analytical charts.

Several experts pointed out that the information articles within the tool may require further refinement. The HCI researcher and the caregiver support organization director noted that the tone and storytelling could be more emotionally sensitive and better aligned with caregivers' lived experiences.

Finally participants also stressed that caregiving situations are highly individual and require flexible design approaches:

> "*I don't believe in one size fits all solution… the situation for the carers are unique.*" (NKA)

Overall, participants' reflections highlighted the importance of clarity, accessibility, emotional sensitivity, and tailored support when designing digital tools intended for caregivers.

4.4.4 Factors Influencing the Adoption of Digital Caregiving Tools

This theme explores the factors that shape whether older caregivers are willing and able to adopt digital support tools. It highlights critical considerations of perceived usefulness, trust, emotional readiness, ease of use amidst caregiver exhaustion, and expectations of system integration.

Experts emphasized that caregivers will only adopt a tool if it provides clear, immediate value in their daily lives. Rather than adopting technology for technology's sake, caregivers look for direct relief. As CR1 explained:

> "*They will only use a tool if they feel it actually helps them in their daily situation.*" (CR1)

Beyond immediate usefulness, trust and credibility emerged as a critical barrier to adoption. Caregivers and the professionals who guide them frequently look for whether a tool is evidence-based and endorsed by healthcare authorities. CR2 highlighted that without proper validation, researchers and clinicians cannot act as advocates for a tool:

> "*We can only recommend tools with proper evidence… otherwise we are not allowed to recommend it.*" (CR2)

This need for simplicity and clinical trust is closely tied to caregivers' limited time and energy. Many older caregivers are already overwhelmed by their daily responsibilities, leaving them with very little cognitive or emotional capacity to learn complex new systems. As CR1 noted:

> "*It cannot be too complex or too time-demanding… many caregivers are overwhelmed.*" (CR1)

This exhaustion is further compounded by the emotional strain of caregiving, which often forces older adults to deprioritize their own health and well-being while focusing entirely on the patient. This self-neglect frequently delays or prevents engagement with digital support tools altogether. CR1 explained this mindset:

> "*They put their needs on pause… and then suddenly they don't know where to turn.*" (CR1)

The caregiver participant described prioritising caregiving over personal needs:

> "*Everything on my personal needs… I put aside.*" (CG1)

Other interviews reinforced that emotional readiness is a major adoption barrier. CR3 emphasized that many caregivers struggle even to acknowledge their own need for support:

> "*Family members must be brave enough to admit they need help.*" (CR3)

A further barrier relates to caregivers' expectations of system integration. Participants noted that caregivers assume digital tools will be embedded within familiar healthcare platforms. When tools operate in isolation or require additional accounts, adoption becomes unlikely. As CR2 stated:

> "*They want it built into existing systems… and that's always a disappointment.*" (CR2)

Both the HCI researcher and the caregiver support organization director highlighted the consequences of this fragmentation. DR2 explained:

> "*It's another tool outside 1177… this fragmentation makes adoption harder.*" (DR2)

The caregiver support organization director added that many caregivers do not even know where to begin searching for support because they do not identify themselves as caregivers:

> "*Many individuals do not recognise themselves as caregivers, and therefore do not search for caregiver-specific support.*" (CC)

This lack of caregiver identity awareness significantly reduces the likelihood that older adults will seek out or adopt digital tools designed for them.

It was also noted that while healthcare professionals see the value of such tools, political prioritization is lacking:

> "*Nurses and doctors see the need… but politicians don't prioritize caregivers.*" (CR3)

Finally, participants emphasised that digital tools should be designed around caregiver needs rather than institutional priorities:

> "*You have to meet the needs of the people, not the needs of the healthcare system*." (NKA)

These findings reveal a complex web of factors spanning usefulness, professional trust, caregiver exhaustion, emotional readiness, identity awareness, and system expectations that shape older caregivers' willingness to engage with digital tools.

In summary, the interviews revealed several key areas shaping older caregivers' experiences, including emotional and practical caregiving demands, usability challenges in digital tools, feedback on proposed design improvements, and factors influencing adoption. Together, these findings provide a descriptive overview of participants' needs, barriers, and preferences, forming the empirical foundation for the subsequent analysis.

5. Analysis

The findings presented in Section 4 show that older caregivers face a dual set of constraints from ageing and caregiving. Notably, ageing introduces cognitive, sensory, emotional, and motor limitations that directly affect how older adults interact with digital systems. At the same time, the caregiving context adds layers of emotional strain, time scarcity, identity disruption, and systemic fragmentation. To further interpret these findings, this section applies a dual-framework analytical strategy. The Senior Technology Acceptance Model (STAM) is used to interpret ageing-related individual constraints. Meanwhile, the NASSS (Non-Adoption, Abandonment, Scale-up, Spread, and Sustainability) framework is used to analyze caregiving-related contextual and

systemic complexity. Applying these frameworks together provides insights about the individual and contextual challenges of older caregivers when engaging with digital tools such as the Next-of-Kin Compass.

5.1 Ageing-Related Factors (STAM Lens)

This section interprets the findings through the lens of the Senior Technology Acceptance Model (STAM), which explains how ageing-related factors shape older adults' engagement with digital health tools. Across the data collected, ageing emerged as a central determinant of technology acceptance, influencing cognitive load, emotional responses, sensory and motor capabilities, and ultimately the perceived fit of the technology within the caregiver's daily life. Thus, the STAM constructs we present below each illuminate distinct mechanisms through which ageing shapes the design requirements for digital tool design aimed at older caregivers.

5.1.1 Gerontechnology Anxiety

Within STAM, gerontechnology anxiety refers to the discomfort, uncertainty, or cognitive strain older adults might experience when interacting with digital systems. According to the results from this study, this anxiety can emerge from a mismatch between system demands and users' emotional and cognitive capacities. The results showed that usability issues such as those found in the heuristic evaluation can be problematic for general users, but particularly so for older users because of age-related considerations such as limited attention and memory (Liu et al., 2021; Chirayus & Nanthaamornphong, 2019).

As discussed in the reflection workshop and in the interviews, for older caregivers, gerotechnology anxiety might be further intensified by the caregiving context. Older caregivers often operate with reduced cognitive bandwidth due to exhaustion, emotional distress, and the constant need to multitask. Under these conditions, even minor usability frictions such as unclear labels, navigation issues, and long text blocks can become disproportionately burdensome. For instance, one researcher highlighted how excessive interaction steps amplify cognitive strain for older users who are already operating under emotional and practical pressure from caregiving:

> "*If it's ten clicks before you get to the information, it's not working*" (CR1).

From a STAM framework perspective, such anxiety signals a structural incompatibility between interface complexity and ageing cognitive profiles. This means that when usability issues are present, such as unpredictable navigation or overwhelming information, the technology is perceived as incompatible with the user's abilities and situational constraints. Partly because of this incompatibility, the interviews highlighted that older adults often "need help to find things" (CC), which is a common pattern: older users usually access digital health assisted by someone (Rasi et al., 2020).

5.1.2 Cognitive Self-Efficacy

Cognitive self-efficacy, which is the users' belief in their ability to successfully operate a digital tool, is a central determinant of technology acceptance in STAM. The interviews indicate that many older caregivers often approach digital systems with low confidence, shaped by unfamiliarity with interface conventions, fear of errors, and difficulty remembering steps. This is consistent with literature showing a correlation between advancing age and a decline in digital confidence (Nymberg et al., 2019; Åberg et al.,

2024). As one researcher explained, illustrating how memory limitations and reduced processing speed directly undermine self-efficacy:

> *"Even if you explain how to use it, next time they may not remember… then it's not a help"* (CR2).

Low cognitive self-efficacy is further compounded by the emotional burden of caregiving, which limits available mental resources for learning and interaction. Under these conditions, digital tools that assume memory retention, exploration, or trial-and-error learning place unrealistic demands on users. During the interviews, the HCI researcher emphasized that even the emotional framing of screening questions can increase cognitive load, making interactions feel heavier and more demanding than intended.

Such low self-efficacy is recognized in STAM to reduce perceived ease of use and increases the likelihood of avoidance or abandonment. This is consistent with Nymberg et al.'s (2019) findings: adoption is hindered by digital confidence decline.

Notably, the findings suggest that many digital systems implicitly assume a level of cognitive continuity and confidence that may not be present among older caregivers. This assumption can be observed in the usability issues found during the heuristic evaluation.

5.1.3 Health Status and Age-Related Functional Decline

STAM emphasizes that sensory, motor, and cognitive declines associated with ageing significantly shape technology use. The interviews provide evidence of these limitations among older caregivers. As one researcher noted:

> "*Many older caregivers may be ill themselves… and they have physical limitations, vision problems, hearing problems, or cognitive issues that make everything harder*." (CR2)

The study findings were consistent with those highlighted by Liu et al. (2021) and Chirayus & Nanthaamornphong (2019), which show that the ageing process negatively affects vision, hearing, motor, and cognitive functions.

The reflection workshop highlighted that several usability issues found during the heuristic evaluation were challenging interactions for older caregivers. For example, extensive scrolling emerged as a potential obstacle, especially when long pages require sustained attention or precise motor control. Similarly, dense text was described as potentially tiring during the interviews, particularly for caregivers already experiencing emotional and physical fatigue. In the reflection workshop and interviews, participants widely supported audio alternatives, noting that listening can be easier than reading for users experiencing visual strain or cognitive overload.

In accordance with the STAM framework, these limitations increase the physical and cognitive effort required to engage with digital tools. When interaction demands exceed the user's functional capacity, the technology becomes effectively inaccessible, regardless of its informational value. This suggests that when digital interfaces are designed around assumptions of stable sensory and cognitive ability, they do not hold in ageing populations.

5.1.4 Synthesis

Taken together, these findings demonstrate that ageing introduces significant constraints that shape the acceptance of digital health tools among older caregivers:

- **Gerontechnology Anxiety**: Complex navigation structures, excessive interaction steps, and information overload intensify cognitive and emotional strain, reducing perceived compatibility with digital systems.
- **Reduced Cognitive Self-Efficacy**: Limited confidence in remembering procedures, understanding interfaces, and recovering from mistakes undermines sustained engagement with digital tools.
- **Functional Accessibility Constraints**: Age-related sensory, motor, and cognitive decline increase interaction effort, particularly when interfaces rely on scrolling, dense text, or visually demanding layouts.
- **Contextual Vulnerability**: The caregiving environment amplifies ageing-related limitations through emotional stress, fatigue, multitasking, and time pressure, making even otherwise usable systems difficult to sustain in practice.

However, while STAM provides a strong explanation of individual-level barriers, it does not fully account for the conditions under which these interactions take place. Our findings show that older caregivers do not engage with digital tools in stable or controlled environments. Instead, they operate under conditions of emotional strain, time pressure, and competing demands. This suggests that even well-designed, age-friendly systems may fail if they do not account for the broader caregiving context in which they are used.

5.2 NASSS Framework: System-Level Complexity in Caregiving Contexts

While STAM explains how ageing shapes individual engagement with digital tools, it does not fully capture the complex environment in which this interaction occurs. The Non-adoption, Abandonment, Scale-up, Spread, and Sustainability (NASSS) framework provides a broader systems lens, revealing interacting layers of complexity that extend beyond individual capabilities. Interpreting the study results through NASSS highlights how different domains of digital health technologies do not simply add complexity but shape the conditions for technology use.

5.2.1 Domain 1: The Condition

Under NASSS, the “condition” can refer both to the clinical diagnosis and to the caregiving experience. For the Next-of-Kin Compass, the relevant condition is caring for a person with cancer. The study interviews show that this condition is unpredictable, as patients’ needs fluctuate and crises emerge without warning. In addition, the interview participants mentioned that caregiving is a profound disruption of identity and life trajectory where caregivers frequently suspend their own needs to prioritize the patient’s. As participants noted in the interviews:

> “*They put their needs on pause….*” (CR1)

> “*Everything on my personal needs… I put aside.*” (CG1)

In addition, it was found in the reflection workshop and interviews that the condition is also accompanied by sustained emotional strain.

This means that caregivers do not engage with systems under relatively stable cognitive and emotional conditions. Instead, in caregiving contexts, interaction is fragmented, emotionally charged, and possibly interrupted. As a result, tools that do not account for this reality risk being perceived as misaligned or irrelevant. This suggests that caregiving-oriented systems must be sensitive to fluctuating emotional and cognitive capacity, rather than assuming stable engagement patterns.

5.2.2 Domain 2: The Technology

This domain considers the material complexity of the technology and the knowledge required to use it. The heuristic evaluation findings in this study showed that the Next-of-Kin Compass was somewhat complex in this domain, with 19 usability barriers identified. As identified in the reflection workshop and the semi-structured interviews, for older caregivers already operating under high cognitive and emotional load, usability issues like those identified might be barriers to adoption. For instance, one interviewee described about existing caregiving digital tools:

> *"The technology has not improved the situation… it has been an added challenge for the person being cared for and an added challenge for the informal caregiver."* (NKA)

To address this, the redesign introduced audio playback, voice-note input, and a short onboarding flow. These changes aimed to mitigate complexity by reducing motor demands and lowering the knowledge-to-use threshold. As described in the interviews, these changes reduced complexity:

> "*Audio options can make information more accessible, especially for users who may find long texts difficult to read." (DR1)*

However, the findings also show that technological complexity is not limited to interface interaction. It also lies in the knowledge the system produces. The reflection workshop and the interview findings suggested the need for knowledge that is actionable and immediately useful, rather than purely informational. At the same time, interviews stressed that trust depends on clinical validity and evidence. Therefore, reducing technological complexity means generating dependable and actionable outputs aligned with caregiver needs and expectations too.

5.2.3 Domain 3: The Value Proposition

This domain evaluates value from both demand-side (caregivers) and supply-side (healthcare organizations) perspectives. From the caregiver perspective, value is closely tied to reducing transaction costs across time, effort, and cognitive load. The findings across the reflection workshop and the interviews indicate that caregivers engage with tools only when they provide immediate and tangible relief.

> "*They will only use a tool if they feel it actually helps them in their daily situation.*" (CR1)

If a system requires sustained cognitive effort without clearly reducing daily burden, perceived value diminishes. On the other hand, from the healthcare organizational side, value is shaped by clinical legitimacy. The interviews found that clinicians generally do not endorse tools without validation, such as clinical trials.

> "*We can only recommend tools with proper evidence… otherwise we are*

not allowed to recommend it." (CR2)

Similarly, the interviews suggest that value is maximized when the tool achieves innovation system fit by integrating into established infrastructures such as 1177, which function as high-trust entry points in the Swedish healthcare system. Without such integration, the perceived value remains limited despite potential benefits for the user.

5.2.4 Domain 4: The Adopter System

Under NASSS, adoption depends on alignment between the technology and the adopter system's (caregivers and healthcare professionals) identities, values, capabilities, and roles. In this study, interviews suggest that a key barrier for caregivers to use self-support tools might be their lack of identification with the "caregiver" label, as this risks a misalignment between the tool's intended function and users' self-perception. In addition, it was found that caregivers often prioritize the patient's needs over their own, which might reduce willingness to engage with tools that contradict this hierarchy of priority. These dynamics challenge the assumption that users will actively seek out supportive technologies, even when such tools may be beneficial. Instead, as interviewees described:

> *"They don't know they are 'next of kin'… they search for 'my husband has cancer,' not 'I need support.'"* (CC)

> "*Many individuals do not recognise themselves as caregivers, and therefore do not search for caregiver-specific support.*" (CC)

The reflection workshop and interview findings also suggested that older caregivers, in particular, often have limited energy to learn new systems, and complexity arises when tools require digital skills or familiarity that are not present. Language barriers can further increase cognitive effort and act as access barriers. For healthcare staff, adoption is shaped by professional norms, workload constraints, and perceived role boundaries. In turn, tools that disrupt established workflows may be perceived as misaligned with professional practice. Consequently, across both groups, adoption is most likely when technologies align with existing identities, capabilities, and routines, using relatable language, minimizing skill requirements, and supporting rather than disrupting established roles and practices.

5.2.5 Domain 5: The Organization

This domain examines a healthcare organization's capacity to integrate an innovation into existing structures. In this case, the study focuses on the possible integration of the Next-of-Kin Compass in the Swedish healthcare system. In reference to it, an interviewee described:

> *"In Sweden currently it is a jungle… many carers describe it as a challenge to navigate the system."* (NKA)

This suggests that adoption of the Next-of-Kin Compass could risk being hindered by system complexity. If there is limited integration with established platforms such as 1177 or standard clinical workflows, this requires additional effort from both staff and users. Implementation, therefore, can require substantial organizational work, including helping staff understand the purpose and value of the tool and embedding it into everyday practice. Therefore, the study findings across the reflection workshop and

interviews suggest that successful adoption of a technology such as the Next-of-Kin Compass is largely dependent on the capacity to integrate it within existing care pathways.

5.2.6 Domain 6: The Wider Context

This domain considers the broader political, regulatory, and institutional environment shaping adoption. Although the study does not exhaustively explore this aspect, interviews suggest that caregiver support is often deprioritized at the policy level, limiting institutional incentives for digital caregiver interventions:

> "*Nurses and doctors see the need… but politicians don't prioritize caregivers.*" (CR3)

Interviewees also mentioned regulatory constraints and fragmented responsibility between healthcare and social services might further complicate implementation. This means that although tools can be appropriately designed for older caregivers, they still might face structural barriers to scaling and long-term sustainability if the wider context is not supportive of them.

5.2.7 Synthesis

Across the NASSS domains, the findings show that complexity is not located in a single layer but emerges through interaction between different domains. As a result, the success of tools like the Next-of-Kin Compass depends not only on interface design but also on contextual alignment.

Thus, the findings identify several system-level dimensions of complexity that shape the adoption and sustainability of digital caregiver support tools:

- **Condition Complexity**: Caregiving for a person with cancer occurs under unstable emotional, cognitive, and temporal conditions, making fragmented and interrupted interaction patterns the norm rather than the exception.
- **Technological Complexity**: Complexity emerges not only from interface usability barriers but also from the need to produce actionable, trustworthy, and clinically credible support that caregivers can immediately apply in practice.
- **Value Alignment Requirements**: Perceived value depends on whether the technology reduces cognitive and practical burden for caregivers while simultaneously meeting healthcare expectations regarding legitimacy, validation, and infrastructural integration.
- **Adopter System Misalignment**: Adoption barriers arise when technologies conflict with caregiver identities, exceed users' digital capacities, or disrupt established professional roles and routines within healthcare practice.
- **Organizational Integration Constraints**: Limited integration with existing healthcare infrastructures and workflows increases implementation burden and reduces the likelihood of sustained organizational adoption.
- **Wider Structural Barriers**: Policy fragmentation, regulatory constraints, and weak institutional prioritization of caregiver support limit scalability and long-term sustainability, even when technologies are well-designed.

This means that effective digital health design for older caregivers requires systemic alignment across user capabilities, caregiving realities, organizational workflows, and institutional structures. Next, having analyzed the challenges related to ageing and

caregiving, the following discussion explores the connections between individual and systemic factors and how those shape digital health design recommendations.

6. Discussion

This section addresses the central research questions guiding this thesis. To achieve this, the section explores significant interactions between the Senior Technology Acceptance Model (STAM) and the Nonadoption, Abandonment, Scale-up, Spread, and Sustainability (NASSS) framework. The findings reveal that there are complex interactions where system complexity and reduced user capacity reinforce each other. In addition, the section outlines the design guidelines that emerge from these interactions and shows their application in the Next-of-Kin Compass digital tool.

6.1 Capacity Depletion from Ageing and Caregiving Challenges

A central finding across the reflection workshop, interviews, and the analytical interpretation of the redesign process is that older caregivers experience a compounded diminishment of cognitive, emotional, and temporal resources. This is aligned with Egwuonwu et al.'s findings (2025), which describe cancer caregiving as involving multidimensional pressure, where emotional, logistical, and physical demands interact and exceed available resources. However, this study found that not only do the dimensions of caregiving interact, but also those from ageing. The result is a unique set of constraints and risks associated with digital engagement. However, each form of depletion operates through a distinct mechanism, shaping how capacity limitations are expressed in interaction with digital health tools.

Cognitive depletion reflects a reduction in processing capacity associated with ageing that is further intensified by the sustained demands of caregiving. Age-related declines in processing speed, working memory, and attentional control reduce baseline cognitive resources available for digital interaction. At the same time, caregiving for individuals with complex conditions such as cancer introduces continuous demands for monitoring, decision-making, and coordination across medical and logistical tasks. These combined pressures increase task switching and reduce the cognitive bandwidth available for engaging with digital health systems, particularly those requiring sustained attention or multi-step interaction sequences. This interaction between cognitive burden and interface demands became especially visible during the heuristic evaluation, where issues such as dense information structures, ambiguous navigation cues, and multi-step interaction flows were identified as likely to increase interpretive and memory-related effort for older caregivers. Workshop discussions further reinforced that these interaction patterns become significantly more burdensome under conditions of caregiving strain.

Meanwhile, temporal depletion arises from the sustained time demands associated with caregiving responsibilities, which reduce the availability of time for digital engagement. Unlike cognitive depletion, this mechanism is primarily structural rather than psychological, reflecting external constraints imposed by continuous caregiving responsibilities. This was strongly reflected in interviews, where caregiving stakeholders and the older caregiver participant described caregiving routines as fragmented, interruption-driven, and unpredictable. These findings directly informed

redesign decisions such as resumable interaction flows with a save-and-finish-later functionality.

Emotional depletion reflects the sustained emotional strain associated with caregiving for individuals with complex and often progressive conditions such as cancer. This includes emotional labor, stress regulation, and continuous concern for the well-being of the care recipient. Over time, these demands contribute to an overall burden, which may reduce available resources for engaging with additional digital tasks. This challenge was particularly evident during interviews and workshop discussions, where participants emphasized that emotionally demanding elements in interfaces, disruptive notifications, and overwhelming information could intensify existing caregiver stress rather than alleviate it. These reflections subsequently informed redesign priorities such as those related to predictability, reduced urgency cues, and calmer interaction flows.

The effect of the described compounded burden is supported by behavioral evidence from prior studies on caregiving intensity and digital engagement. Riegel et al. (2025) demonstrate that increased caregiving intensity is associated with a lower likelihood of successful engagement with a digital intervention, while Yunis et al. (2024) report reduced adherence to high-frequency reporting tasks. Although these studies do not directly measure cognitive load or time availability, their findings are consistent with the proposed mechanism in which sustained caregiving demands reduce the capacity to initiate and maintain engagement with cognitively demanding digital systems.

However, beyond capacity depletion, the findings also suggest a pattern in how caregivers allocate engagement across different types of digital health tasks. Sustained caregiving burden appears to be associated with deprioritization of personal well-being, which may reduce engagement with self-supportive digital tools. Under conditions of limited cognitive and temporal capacity, caregivers appear to allocate available resources primarily toward patient-related responsibilities rather than their own needs. This pattern emerged strongly during interviews with caregiving stakeholders and was reflected in workshop discussions concerning the perceived relevance of caregiver-focused features. This was also found by Riegel et al. (2025), who observed that highly burdened caregivers were often unable to devote even minimal time toward their own well-being, and by Yunis et al. (2024), which found higher adherence when reporting patient symptoms compared to self-focused monitoring. Together, these findings indicate a consistent asymmetry in engagement across task types under caregiving burden.

Importantly, this pattern suggests that engagement with caregiver-support technologies is shaped not only by capacity constraints but also by the evaluation of the perceived relevance and legitimacy of self-directed care. Within the STAM framework, perceived usefulness is, therefore, not a fixed property of the system but a judgment of effort, which is shaped by constrained cognitive, temporal, and emotional resources for older caregivers. This means that functionally beneficial tools may be deprioritized if the interaction effort required is perceived as disproportionate relative to caregiving demands. Even more, their relevance might be perceived as limited if it is not aligned with the identity and priorities of the user.

6.2 System Amplification of Burden

System-level conditions amplify the effects of reduced capacity by increasing coordination demands, interaction complexity, and cognitive switching requirements.

This insight emerged across the heuristic evaluation, workshop discussions, stakeholder interviews, and redesign process. This amplification occurs through two interrelated mechanisms: structural fragmentation across digital health systems and design-level mismatches that increase interaction effort under conditions of reduced cognitive and motor capacity.

First, fragmentation across healthcare digital systems requires caregivers to navigate multiple interfaces, accounts, and information sources. This increases coordination overhead and cognitive switching demands, thereby compounding the effects of reduced cognitive capacity associated with ageing and caregiving strain. In practical terms, fragmented systems disrupt interaction continuity by requiring users to repeatedly shift between platforms, reconstruct task context, and manage distributed information flows. These challenges were repeatedly highlighted during stakeholder interviews, where caregiving researchers and support organisation representatives described caregivers as frequently acting as coordinators across disconnected services and information channels. In contrast, institutional embedding within established healthcare infrastructures reduces this burden by stabilizing access pathways and consolidating interaction routines. Findings from Yunis et al. (2024) suggest that mobile health tools are most effective when they integrate smoothly into users' existing routines. Where such integration is absent, even minor usability issues can accumulate into significant interaction costs due to repeated disruption of established routines.

Importantly, system design itself can actively produce additional burden when it fails to accommodate age-related cognitive, sensory, and motor changes. Reduced working memory capacity, slower processing speed, sensory decline, and diminished motor coordination increase vulnerability to interfaces that rely on high cognitive interpretation or precise interaction. This aligns with Alsswey et al. (2022) and Liu et al. (2021), who emphasize that many mobile health applications insufficiently account for perceptual, motor, and cognitive constraints among older users. Similarly, prior research suggests that interaction difficulties are more strongly associated with interface-induced cognitive and motor load than with chronological age itself (Gomez-Hernandez et al., 2023; Khamaj & Ali, 2024; Morey et al., 2019; Van Acker et al., 2023). Within the present study, these issues were concretely identified during the heuristic evaluation through observations related to ambiguous interaction cues, insufficient guidance, and reliance on memory or exploration. Workshop discussions further reinforced how such interaction patterns become disproportionately demanding within caregiving contexts characterised by fatigue, stress, and divided attention.

The present findings extend this literature by demonstrating how specific design choices can translate into cumulative interaction costs. For example, ambiguous labels such as "Read more" require users to infer system outcomes, thereby increasing uncertainty and cognitive effort, whereas explicit labels reduce interpretive load and support more efficient interaction. These findings emerged directly from the heuristic evaluation and were subsequently discussed during the reflection workshop in relation to ageing-related declines in cognitive processing and confidence. During redesign, these observations informed efforts to increase interaction clarity through explicit affordances, simplified navigation pathways, and progressive disclosure structures. These micro-level design decisions accumulate across interaction sequences and can cause disproportionate burden under conditions of reduced cognitive capacity. This indicates that interaction burden is not solely an outcome of caregiving strain or ageing-related decline but is also actively produced through interface design when system demands exceed user capacity.

This same mechanism also shapes perceived self-efficacy during interaction. Interview findings and workshop reflections suggested that when interfaces require sustained interpretation, memory retrieval, or precise motor input, users can experience reduced perceived control over system outcomes. Conversely, clear affordances, explicit cues, and consistent navigation structures support more predictable interaction pathways, thereby strengthening confidence and reducing avoidance behavior. In this sense, self-efficacy is not just a trait of an individual. Instead, it is mediated by systemic complexity the broader socio-technical environment and technology design.

6.3 Capacity Versus System Complexity

Overall, engagement seems to be governed by interactions between capacity depletion and system complexity. This interpretation emerged through the synthesis of results across the heuristic evaluation, reflection workshop, stakeholder interviews, and iterative redesign process. While the heuristic evaluation identified concrete usability barriers, interviews and workshop discussions demonstrated how these barriers become amplified under conditions of ageing and caregiving. The redesign process further reinforced this relationship by illustrating that isolated usability improvements were often insufficient unless broader interaction flows and caregiving realities were also addressed simultaneously.

In a nutshell, when capacity is severely constrained and systems are highly complex, non-adoption becomes more likely. This relationship became especially visible when findings from the heuristic evaluation were interpreted through the NASSS and STAM frameworks. The emerging patterns indicate that digital health engagement is structured by a dynamic in which capacity and complexity interact multiplicatively. This interpretation is supported by meta-analytic evidence showing that more complex digital interventions are associated with higher dropout rates among older caregivers (Tan et al., 2025). However, it should be noted that as this dynamic is inferred from converging qualitative sources rather than directly measured, its strength remains indicative rather than causal. Even so, a key implication is that design guidelines must address cumulative burden across interaction design, emotional load, and system coordination rather than focusing on isolated interface optimization.

6.4 Design Guidelines

The preceding discussion demonstrated that digital engagement among older caregivers is shaped by the interaction between ageing-related capacity constraints and system-level caregiving complexity. Across the heuristic evaluation, reflection workshop, stakeholder interviews, and iterative redesign process, the findings consistently indicated that burden arises from the interaction between reduced cognitive capacity, constrained temporal resources, emotional strain, and system-level complexity.

Consequently, from the synthesis of all findings, the study proposes design guidelines that address the challenges found. While existing age-friendly design guidelines primarily focus on accessibility, usability, and age-related sensory, motor, and cognitive changes among older adults (Liu et al., 2021; WCAG, 2018), the guidelines proposed in this study extend prior work by explicitly addressing the compounded burden arising from the intersection of ageing and caregiving. In particular, they emphasise temporal scarcity, emotional burden, patient-centred prioritisation, and system-level

fragmentation as key determinants of digital engagement among older caregivers. Together, these dimensions contribute a capacity-demand alignment perspective that is not typically integrated within age-friendly design frameworks. The following are the design guidelines:

6.4.1 Minimising Cumulative Cognitive Load

Cognitive capacity is reduced in older caregivers due to age-related changes in processing speed, working memory, and attentional control and is further constrained by the sustained demands of caregiving. Importantly, these limitations do not only affect discrete interaction steps but accumulate across repeated engagement with digital systems, particularly where tasks require multi-step navigation, recall of prior actions, or sustained attention. Cognitive burden is therefore best understood as a sequential and compounding phenomenon rather than an isolated usability issue.

Design should therefore aim to minimize cumulative cognitive load by structuring interactions in ways that reduce memory dependence, decision complexity, and interpretive effort over time. Consequently, strategies such as progressive disclosure, chunking of information, explicit labeling, and reduction of choice complexity (Mishra et al., 2025) function as mechanisms for aligning system demands with constrained cognitive capacity. These approaches aim to prevent the accumulation of cognitive effort across interactions.

6.4.2 Designing for Temporal Scarcity

Temporal constraints represent a structural condition of caregiving rather than a situational limitation. Caregivers frequently engage with digital systems in fragmented intervals, interrupted by competing responsibilities and fluctuating caregiving demands. Under these conditions, interaction breakdown is attributable to incompatibility between system design and the temporal realities of caregiving.

Design must therefore assume intermittent use as the default engagement pattern. This requires support for pause and resume continuity without loss of progress. minimization of mandatory sequential steps, reduction of redundant interaction requirements, and embedding of tasks within existing caregiving routines (Johnson et al., 2022; Yehuda et al., 2024). These practices are preconditions for feasible engagement for the older caregiver more than optimization fixes only.

6.4.3 Managing Emotional Burden Through Interaction Containment

Emotional strain in caregiving reflects a complex condition of responsibility, stress, and concern for the care recipient. This reduces tolerance for additional cognitive and interactional demands and means that emotionally burdensome design can directly contribute to disengagement. Importantly, emotional burden is influenced by content (which can be inherently triggering because it relates to the illness itself) but also by the structure and predictability of interaction itself.

Design implications extend beyond affective tone to the containment of interaction-induced burden. Systems should minimize unnecessary alerts, avoid urgency amplification, reduce avoidable decision points, and maintain stable and predictable interaction flows across tasks. Interaction predictability reduces perceived

effort and limits escalation of emotional strain during use (Giger et al., 2026; Tan et al., 2025). Emotional design, in this sense, functions as a mechanism for regulating interaction-induced load rather than only being an aesthetic or communicative enhancement.

6.4.4 Supporting Patient-Centered Perspectives

The findings indicate that caregivers prioritize patient-related tasks over self-directed support in the context of cancer caregiving. Design must therefore account for this explicitly. Self-support functionalities should not require separate initiation or additional cognitive effort but should be embedded within patient-related workflows and contextual points of relevance for the user. Reducing interaction cost is therefore critical, as self-directed tools must operate within, rather than compete against, dominant priorities. This implies an integration of caregiver-support functions into existing task flows rather than parallel system architectures.

6.4.5 Reducing System-Level Fragmentation and Coordination Overhead

System fragmentation across healthcare platforms increases cognitive switching demands and disrupts interaction continuity. Each transition between systems requires users to reconstruct context, reorient tasks, and manage distributed information sources. These demands are amplified under conditions of reduced cognitive capacity, where switching costs become disproportionately burdensome.

Design should therefore aim to reduce fragmentation through infrastructural alignment and consolidation of interaction pathways. Where possible, embedding within established healthcare systems such as 1177 reduces coordination burden. Where full integration is not feasible, consistency across platforms and minimisation of cross-system transitions become essential design requirements. The central design objective is therefore the reduction of context reconstruction effort across distributed systems.

6.4.6 Constructing Self-Efficacy Through Interaction Design

Self-efficacy is partially constructed through interaction design and is not solely an individual-level psychological attribute. The findings indicate that perceived control over system use is shaped by the clarity, predictability, and transparency of interface behavior. Interfaces that provide explicit affordances, consistent navigation structures, and immediate feedback enhance perceived control by reducing interpretive uncertainty. Conversely, ambiguous cues and inconsistent interaction logic might increase cognitive effort and reduce confidence. Therefore, supporting self-efficacy requires reducing interpretive burden, ensuring transparency of system states, and maintaining consistent interaction logic across tasks and contexts. In this sense, confidence in system use is produced through the predictability of interaction rather than prior experience alone.

6.4.7 Accommodating Multimodal Age-Related Accessibility Constraints

Age-related changes in sensory, motor, and cognitive functioning increase vulnerability to interfaces that rely on precision, memory, or interpretation. Design should therefore incorporate accessibility adaptations that reduce physical and cognitive effort. This includes large interactive elements, high-contrast visual design, and simplified navigation structures that reduce reliance on recall. Where appropriate, multimodal interaction options such as voice input or audio output can further reduce interaction demands. These adaptations should be treated as essential accessibility requirements rather than optional enhancements, as they directly determine whether interaction is feasible for users with different functional capacities (Liu et al., 2021).

6.4.8 Integrative Principle: Capacity-Demand Alignment

Across all stages of this research, the findings converge on a main design requirement: digital health systems must align interaction demands with cumulative user burden across cognitive, temporal, emotional, and infrastructural dimensions. The primary design challenge is not optimization of isolated usability features but reduction of compounded effort across interaction sequences and system contexts. Effective design needs to ensure that interaction demands do not exceed the constrained and unique capacities of older caregivers. The following table summarizes each of the design guidelines described so far:

Table 3. Summary of Design Guidelines

Design Guideline	**Description**	**Example Design Strategies**
Minimise cumulative cognitive load	Reduce cognitive burden across interaction sequences by lowering memory dependence, decision complexity, and interpretive effort.	Simplified navigation structures; progressive disclosure; persistent on-screen guidance; limiting the number of choices per screen.
Design for temporal scarcity	Support fragmented, intermittent use by reducing dependency on uninterrupted, sequential task completion.	Autosave functionality; resumable task flows; short interaction sequences; continuity-preserving reminders.
Contain emotional burden	Reduce interaction-induced emotional strain by ensuring predictable and non-escalating interaction flows.	Calm and neutral notification language; consistent screen layouts; minimised use of disruptive pop-ups or urgent alerts.

Support patient-centered perspectives	Reduce interaction cost by integrating caregiver-support functions directly into patient-centred tasks.	Stress-management prompts embedded in medication or appointment flows; contextual caregiver check-ins within care tasks.
Reduce system-level fragmentation	Decrease coordination burden by aligning with existing healthcare infrastructures and limiting cross-platform transitions.	Consolidated dashboards; consistent navigation patterns across services; integration with established systems such as 1177.
Support self-efficacy through interaction design	Enhance perceived control through clear, predictable, and transparent system behaviour.	Clear affordances and labels; consistent navigation structures; visible progress indicators; immediate confirmation feedback after actions.
Accommodate multimodal accessibility constraints	Address sensory, motor, and cognitive limitations through accessible and flexible interaction modalities.	Large touch targets; high-contrast visual design; adjustable text size; voice input; audio-supported instructions.

6.4.9 Design Application

Having established how ageing and caregiving challenges shape digital engagement (RQ1 and RQ2), this section addresses RQ3 by examining how these insights can be operationalised in a real-world application. As described in the methodology, the design guidelines were applied to the redesign of the Next-of-Kin Compass. Through an Annotated Portfolio, the section provides a visual and analytic account of how each guideline informs specific interface changes intended to target compounded challenges of the older caregiver. In doing so, this section illustrates how theoretically and empirically grounded insights can be transformed into actionable design improvements that align system demands with the constrained capacities of the intended user group.

The following three annotated screens (Figures 16-18) show how the guidelines were applied. Each links to a specific guideline from the summary table.

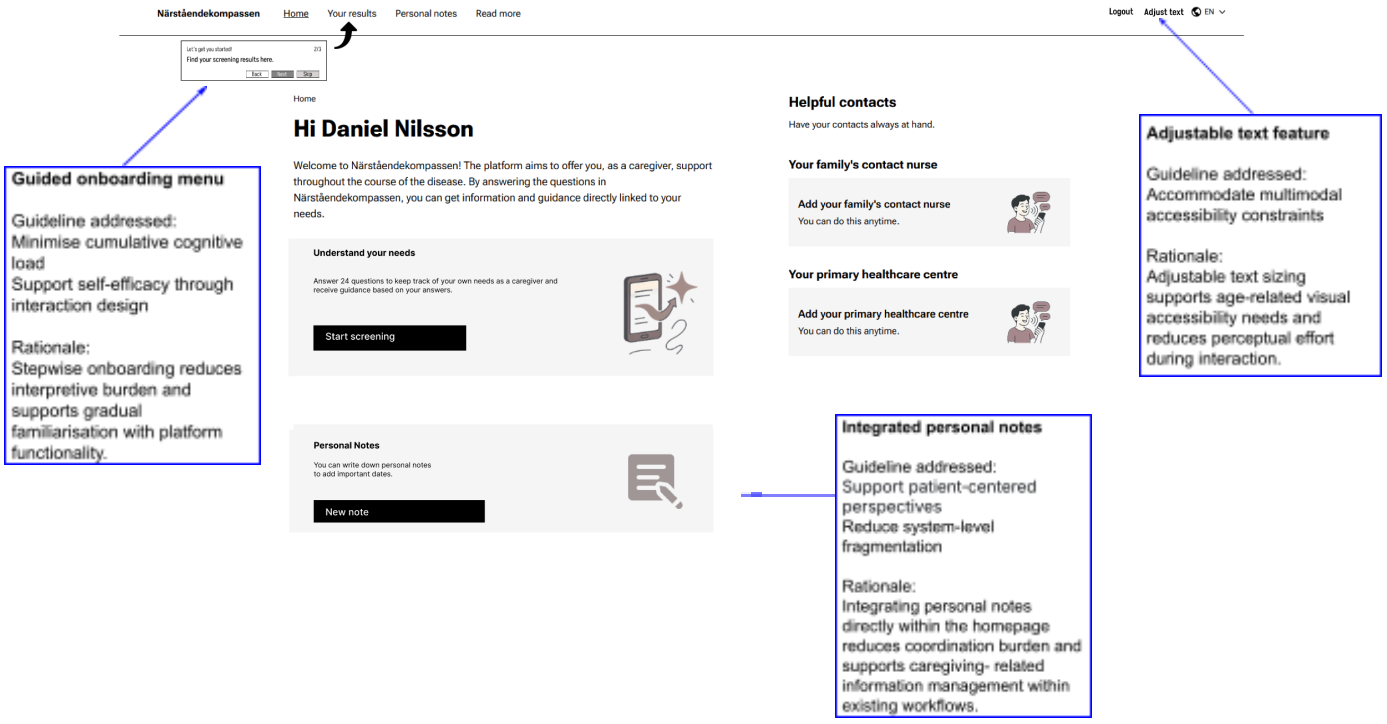


Figure 16. Annotated redesign of the Next-of-Kin Compass homepage, showing the guided onboarding menu (Minimise cumulative cognitive load, Support self-efficacy), adjustable text feature (Accommodate multimodal accessibility), and integrated personal notes (Support patient-centered perspectives, Reduce system-level fragmentation), each linked to the corresponding design guidelines.



Figure 17. Annotated redesign showing the "Save and finish later" interaction (Design for temporal scarcity) linked to the relevant design guidelines.

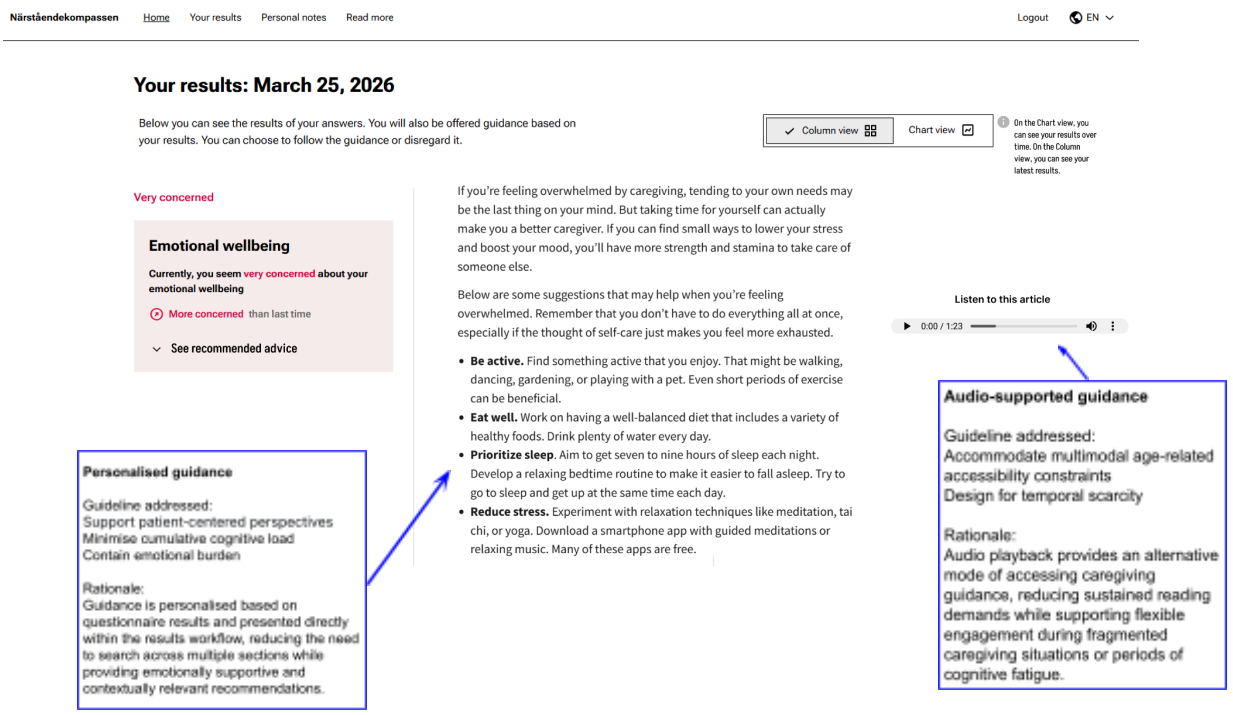


Figure 18. Annotated redesign showing audio-supported input (Accommodate multimodal age-related accessibility, Design for temporal scarcity), and progressive disclosure elements and personalised content (Support patient-centered perspectives, Minimise cumulative cognitive load, Contain emotional burden) linked to the relevant design guidelines.

As shown in Figures 16-18, the annotated portfolio demonstrates how the design guidelines were concretely put into practice in the context of the Next-of-Kin Compass. The visual annotations make explicit how each interface decision responds to specific challenges, showing a direct line from empirical findings to design rationale. Overall, this serves as a practical proof of concept for aligning system demands with the constrained capacities of older caregivers. In the Appendix, a handout of the design guidelines can be found for practical implementation.

The following section synthesizes the study's key contributions and implications, addresses its limitations, and identifies avenues for future research.

7. Contributions and Conclusion

7.1 Theoretical Contribution

This study advances the digital health and human–computer interaction literature by reconceptualizing usability in later life as a context-dependent phenomenon influenced by caregiving responsibilities. Although previous research has identified age-related barriers to technology use (Liu et al., 2021; Denecke et al., 2025), these barriers have primarily been examined at the individual level. The present findings indicate that individual-level approaches are insufficient when technologies are deployed within caregiving contexts.

A primary contribution of this study is the development of a dual-burden perspective,

which conceptualizes usability as the outcome of interacting constraints related to both ageing and caregiving. This perspective extends existing research on caregiving burden (Åberg et al., 2024; Ahmad et al., 2023) by demonstrating that burden not only arises from caregiving responsibilities but also actively shapes how digital technologies are perceived, utilized, and maintained over time.

Furthermore, the study identifies a capacity-demand mismatch as a key theoretical mechanism underlying non-adoption. Digital health tools often impose cognitive, emotional, and organizational demands that exceed the capacity of older caregivers. This finding shifts the focus from static usability barriers to a dynamic understanding of interaction, in which engagement is shaped by fluctuating user resources and contextual pressures.

Additionally, the study offers a methodological contribution by demonstrating the value of integrating the Senior Technology Acceptance Model (STAM) with the NASSS framework (Greenhalgh et al., 2017). While STAM addresses individual-level acceptance factors, NASSS captures system-level complexity. Their integration yields a more comprehensive framework for understanding digital health adoption, illustrating that non-adoption results from the interaction between individual capacity and system demand, rather than from either dimension in isolation. This combined perspective enhances the explanation of real-world adoption challenges by linking personal constraints with systemic conditions.

7.2 Implications for Design Practice

The findings suggest that designing digital health tools for older caregivers necessitates a shift from focusing exclusively on age-related usability issues to addressing the intersection of ageing and caregiving constraints. First, digital systems should be designed for users with limited and fluctuating capacity, rather than for idealized users. Many older adults encounter barriers related to usability, cognitive load, and digital literacy (Liu et al., 2021; Denecke et al., 2025), and the present study suggests that caregiving responsibilities further diminish available capacity. Consequently, even minor usability challenges can substantially impede interaction. Design strategies should therefore prioritize reducing complexity, minimizing interaction steps, and supporting intuitive navigation, consistent with established usability principles (Nielsen, 1994).

Second, emotional factors are integral to usability in caregiving contexts. Caregiving involves stress, responsibility, and emotional burden (Åberg et al., 2024; Ahmad et al., 2023), which directly influence user engagement with digital tools. Accordingly, systems should promote clarity, reassurance, and predictability to enable effective interaction under stressful conditions.

Finally, personalization can reduce interaction effort and enhances accessibility. Given the diversity of users' needs, capabilities, and caregiving situations, adaptable features can align system demands with user capacity. This approach facilitates more flexible and inclusive interaction, especially in contexts where user needs might evolve over time.

7.3 Implications for Healthcare Organizations

For healthcare organizations, the findings underscore that successful adoption of digital health tools depends on their integration within existing systems and practices, rather

than on usability alone. The NASSS framework demonstrates that adoption is influenced by interacting factors, including organizational context and system complexity (Greenhalgh et al., 2017). Consistent with this, the findings indicate that digital tools must be embedded within existing healthcare infrastructures and workflows. Tools operating outside established systems risk increasing coordination burden and diminishing practical usability for both caregivers and healthcare professionals.

Additionally, professional endorsement is a significant factor in adoption. Caregivers frequently rely on healthcare providers as trusted sources of information and are more likely to engage with tools recommended within formal care contexts. This finding reinforces the importance of aligning digital tools with clinical practices and ensuring their credibility within healthcare systems.

Finally, digital tools must be designed to integrate seamlessly into real-world workflows, rather than introducing additional complexity. Systems that require substantial changes to existing routines or increase workload are unlikely to be adopted or sustained, regardless of their potential benefits.

7.4 Implications for Policy

At the policy level, the study highlights the necessity of explicitly recognizing and supporting the role of caregivers within digital health ecosystems. First, caregivers should be recognized as a key target group within digital health strategies. Existing research highlights the substantial responsibilities and burdens associated with caregiving (Åberg et al., 2024; Ahmad et al., 2023), yet digital health initiatives frequently prioritize patients. The present study demonstrates that caregivers play a central role in technology use and should therefore receive direct support.

Second, policy should support the development of accessible and preventive digital tools that alleviate caregiver burden. The findings indicate that tools are more likely to be adopted when they deliver clear, immediate value and reduce daily strain, rather than introducing further complexity.

Finally, policymakers should prioritize the development of trusted and well-integrated digital health platforms. As emphasized by the NASSS framework, successful implementation depends on alignment across technological, organizational, and system-level factors (Greenhalgh et al., 2017). Policy support for integration, validation, and accessibility is therefore essential for enabling adoption and ensuring long-term sustainability.

7.5 Limitations

This study has several methodological and conceptual limitations that should be considered when interpreting the findings. First, the empirical basis of the study is derived from relatively limited qualitative expert-led data. While the study’s approach uses multiple methods that strengthen analytical triangulation, it does not capture the full variability of experiences among older caregivers. In particular, the absence of direct, in-situ observational data from a larger population of older caregivers means that findings are interpreted through mediated accounts rather than sustained longitudinal engagement with end-users in their natural contexts. Although the expert-led perspectives provide valuable systemic and design-oriented insights, they may introduce interpretive bias and might not fully align with users' experiences.

Second, the study adopts a dual-theoretical framing combining STAM and NASSS. While this approach enables multi-level analysis of individual and system-level constraints, it also introduces conceptual overlap and limits the ability to isolate discrete causal mechanisms. The analysis of results, therefore, remains interpretive.

Third, the findings are contextually embedded within the Swedish healthcare environment. As a result, system-level constraints identified in the NASSS analysis may not generalize directly to other healthcare systems.

Finally, the study focuses primarily on usability and adoption-related dimensions of digital health tools. It does not empirically evaluate long-term clinical outcomes, sustained behavioral change, or health impacts. Consequently, conclusions remain bounded to the feasibility of interactions and potential perceived value rather than measured effectiveness.

7.6 Future Work

Future research should extend and refine the study's findings by addressing both empirical and methodological gaps identified in this study. Importantly, there is a need for longitudinal, in-situ studies that examine how older caregivers interact with digital health tools over extended periods of time to have a more grounded understanding of how capacity depletion and system complexity evolve dynamically, particularly considering the illness trajectory and how caregivers adapt to it.

Similarly, direct participatory research with older caregivers as primary design partners is needed. While this study integrates expert and stakeholder perspectives, co-design methodologies would allow for more accurate representation of lived experience and may reveal latent usability strategies that are not visible through expert interpretation alone.

In addition, the dual-framework integration of STAM and NASSS could be further operationalized into a unified analytical model that explicitly maps interactions between individual capacity constraints and system-level complexity factors. This would strengthen the explanatory precision of the dual-burden effect and support a systematic application of this model in design and evaluation contexts.

Moreover, a quantitative validation of the proposed capacity–demand mismatch mechanism is needed. Future studies could operationalize constructs such as cognitive load, emotional burden, interaction cost, and system fragmentation to empirically test their combined effect on engagement, abandonment, and sustained use. This would move the current explanatory model from interpretive synthesis toward a measurable model.

Finally, broader comparative studies across different healthcare systems are required to assess the transferability of the findings. Examining how structural factors such as digital infrastructure maturity, policy support for caregivers, and integration of health platforms influence adoption would clarify the boundaries of the findings.

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9. Appendix

9.1 Interview Guides

This section provides the questions used for each of the semi-structured interviews as guides.

9.1.1 Care Researcher

Introduction

- The focus of this interview is on deepening our understanding of the challenges of caregivers, especially those who are older adults, when using digital tools of support such as Next-of-Kin Compass.

Field of expertise

- What is your field of expertise?

Caregiving challenges (in cancer care)

- How much knowledge or experience do you have with older caregivers?
- What are the challenges of caregiving?
    - What do caregivers need?
    - What does self-care mean in this context?

Older caregivers' challenges (in cancer care)

- How prevalent are older adults in caregiving?
- What specific challenges do older adults face as caregivers?

Dealing with those challenges

- What resources/tools do caregivers have/use for dealing with these challenges?
    - (If they don't mention digital tools, introduce the topic.)
        - Do you know about any digital tools that support caregivers?
- To what extent can digital tools help support caregivers and how?
    - Are there needs that can not be fulfilled with digital tools?
    - What are the other factors that should be considered?

Usability challenges

- How do you measure the effectiveness of these resources/tools?
    - How about the effectiveness of digital tools?
- Do you know about challenges that arise when using these tools?
    - Prompts: time, effort, frustration, anxiety

The current tool

- When talking about Next-of-Kin Compass, how effective do you think it is (for older carers), and why?
- Do you know about challenges that arise when using it?
    - Prompts: time, effort, frustration, anxiety

The updated tool

- In this tool, users can get information and advice on different needs through the "Read More" label. What do you think about this label? Is it helpful for (older) caregivers?
- What do you think about personalizing information and links so that they are specific to the screening results? Is it helpful for (older) caregivers?
- What do you think about including audio alternatives for longer texts? Is it helpful for (older) caregivers?
- What do you think about allowing users to add notes by recording their voice? (Not removing the text option, but adding this feature) Is it helpful for (older) caregivers?

- What do you think about onboarding users in this application? Is it helpful for (older) caregivers?
    - What would you recommend?

Adoption

- What influences the adoption of a tool like this one?

9.1.2 Digital Health Researcher

Introduction

- The focus of this interview is on deepening our understanding of usability, design guidelines, and factors that influence adoption in the context of digital tools for supporting caregivers.
- What is your field of expertise?

Designing digital health tools

- What is your experience in designing and researching digital health tools?
- What is the process when designing digital health tools?
    - How to identify the needs that can be translated into a digital tool?

Designing tools for (older) caregivers

- How much knowledge or experience do you have with designing tools for older users?
- How much knowledge or experience do you have with designing tools for caregivers?
    - How much knowledge or experience do you have with designing tools for older caregivers?
- What design considerations are important when designing tools for older users?
- What design considerations are important when designing tools for caregivers?
    - What design considerations are important when designing tools for older caregivers?

Dealing with those challenges

- What digital resources/tools do caregivers have/use for dealing with challenges they might face?
- To what extent can digital tools help support caregivers and how?
    - Are there needs that can not be fulfilled with digital tools?
    - What are the other factors that should be considered?

Usability challenges

- How do you measure the effectiveness of these resources/tools?
    - How about the effectiveness of digital tools?
- Do you know about challenges that arise when using these tools?
    - Prompts: time, effort, frustration, anxiety

The current tool

- When talking about Next-of-Kin Compass, how effective do you think it is (for older carers), and why?
- Do you know about challenges that arise when using it?
    - Prompts: time, effort, frustration, anxiety

The updated tool

- In this tool, users can get information and advice on different needs through the “Read More” label. What do you think about this label? Is it helpful for (older) caregivers? - Maybe we don't need to ask this question. We already know, haha
- What do you think about personalizing information and links so that they are specific to the screening results? Is it helpful for (older) caregivers?
- What do you think about including audio alternatives for longer texts? Is it helpful for (older) caregivers?

- What do you think about allowing users to add notes by recording their voice? (Not removing the text option, but adding this feature) Is it helpful for (older) caregivers?
- What do you think about onboarding users in this application? Is it helpful for (older) caregivers?
    - What would you recommend?

Adoption

- What influences the adoption of a tool like this one?

9.1.3 Representative of a Caregiving Support Organization

Introduction

- The focus of this interview is on deepening our understanding of the challenges of caregivers, especially those who are older adults, when using digital tools of support such as Next-of-Kin compass.

Field of expertise

- What is your field of expertise?

Caregiving challenges (in cancer care)

- How much knowledge or experience do you have with older caregivers?
- What are the challenges of caregiving?
    - What do caregivers need?
    - What does self-care mean in this context?

Older caregivers' challenges (in cancer care)

- How prevalent are older adults in caregiving?
- What specific challenges do older adults face as caregivers?

Dealing with those challenges

- What resources/tools do caregivers have/use for dealing with these challenges?
    - (If they don't mention digital tools, introduce the topic.)
        - Do you know about any digital tools that support caregivers?
- To what extent can digital tools help support caregivers and how?
    - Are there needs that can not be fulfilled with digital tools?
    - What are the other factors that should be considered?

Usability challenges

- How do you measure the effectiveness of these resources/tools?
    - How about the effectiveness of digital tools?
- Do you know about challenges that arise when using these tools?
    - Prompts: time, effort, frustration, anxiety

The current tool

- When talking about Next-of-Kin compass, how effective do you think it is (for older carers), and why?
- Do you know about challenges that arise when using it?
    - Prompts: time, effort, frustration, anxiety

The updated tool -

- In this tool, users can get information and advice on different needs through the "Read More" label. What do you think about this label? Is it helpful for (older) caregivers?
- What do you think about personalizing information and links so that they are specific to the screening results? Is it helpful for (older) caregivers?
- What do you think about including audio alternatives for longer texts? Is it helpful for (older) caregivers?
- What do you think about allowing users to add notes by recording their voice? (Not removing the text option, but adding this feature) Is it helpful for (older) caregivers?

- What do you think about onboarding users in this application? Is it helpful for (older) caregivers?
    - What would you recommend?
- How could you integrate this tool into their support?
    - Have you considered this as part of your system?

Adoption

- What influences the adoption of a tool like this one?

9.1.4 Older Caregiver

Introduction

- The focus of this interview is on deepening our understanding of the challenges of caregivers, especially those who are older adults, when using digital tools of support such as Next-of-Kin Compass.

Caregiving and ageing challenges (in cancer care)

- What are the challenges you face?
    - What do you need as a caregiver?
    - What does self-care mean for you?

Dealing with those challenges.

- What resources/tools do you have/use for dealing with these challenges?
    - (If they don't mention digital tools, introduce the topic.)
        - Do you know about any digital tools that support caregivers?
- To what extent do you think digital tools can help support you as a caregiver and how?
    - Are there needs that can not be fulfilled with digital tools?
    - What are the other factors that should be considered?
    - What do you need for a tool to be useful and easy to use?

Usability challenges

- What challenges, if any, arise when using these tools?
    - Prompts: time, effort, frustration, anxiety

The updated tool

- How important is it for you to have explicit or direct language in the app?
- How important is it for you to have personalized/targeted information for your specific situation and concerns?
- How important is it for you to be able to listen to long texts instead of reading them in an app?
- How important is it for you to be able to use your voice instead of writing text in an app?
- How important is it for you to have step-by-step instructions when using an app?

Adoption

- What influences the adoption of tools for you?

9.1.5 UX designer

Introduction

- The focus of this interview is on deepening our understanding of usability, design guidelines, and factors that influence adoption in the context of digital tools for supporting caregivers.
- What is your field of expertise?

Designing digital health tools

- What is your experience in designing and researching digital health tools?
- What is the process when designing digital health tools?

    - How to identify the needs that can be translated into a digital tool?

Designing tools for (older) caregivers

- How much knowledge or experience do you have with designing tools for older users?
- How much knowledge or experience do you have with designing tools for caregivers?
    - How much knowledge or experience do you have with designing tools for older caregivers?
- What design considerations are important when designing tools for older users?
- What design considerations are important when designing tools for caregivers?
    - What design considerations are important when designing tools for older caregivers?

Usability challenges

- How do you measure the effectiveness of these resources/tools?
    - How about the effectiveness of digital tools?
- Do you know about challenges that arise when using these tools?
    - Prompts: time, effort, frustration, anxiety

The current tool

- When talking about Next-of-Kin Compass, how effective do you think it is (for older carers), and why?
- Do you know about challenges that arise when using it?
    - Prompts: time, effort, frustration, anxiety

The updated tool (show the power point)

- In this tool, users can get information and advice on different needs through the “Read More” label. What do you think about this label? Is it helpful for (older) caregivers?
- What do you think about personalizing information and links so that they are specific to the screening results? Is it helpful for (older) caregivers?
- What do you think about including audio alternatives for longer texts? Is it helpful for (older) caregivers?
- What do you think about allowing users to add notes by recording their voice? (Not removing the text option, but adding this feature) Is it helpful for (older) caregivers?
- What do you think about onboarding users in this application? Is it helpful for (older) caregivers?
    - What would you recommend?

Adoption

- What influences the adoption of a tool like this one?

9.2 Heuristic Evaluation

Table 2. Summary of Usability Issues Identified via Heuristic Evaluation Adjusted for Older Adults (65+).

Heuristic	**Age-related design criteria**	**Observed issue**
1. Visibility of System Status	Support for limited attention and memory to reduce uncertainty (Liu et	**Sign-up:** No message anticipating that a confirmation email will be

	al., 2021; Chirayus & Nanthaamornphong, 2019).	sent after clicking “Submit.”
		Reminders: Users cannot see or edit the specific time set for receiving email reminders.
2. Match Between System & Real World	Clear labeling and simple language to improve comprehension and reduce cognitive burden (Liu et al., 2021; Morey et al., 2019; Gomez-Hernandez et al., 2023).	**“Concerned” scale:** Users may struggle to distinguish between the options (e.g., “Moderately” vs. “Seriously”).
		Graph views: Users may not know what to expect from the “Column” and “Chart” views.
3. User Control & Freedom	Flat navigation and persistent return buttons to manage motor limitations and accidental actions (Liu et al., 2021; Gomez-Hernandez et al., 2023).	**Screening:** No explicit buttons to save, cancel, or reset a screening in progress.
		Editing: Users must go through every step to submit changes.
4. Consistency & Standards	Consistent terminology and visual cues (color/styles) to reduce cognitive load (Liu et al., 2021; Tajudeen et al., 2022).	**Terminology:** Inconsistent use of “Survey” and “Screening” across the application.
		Navigation: The latest result disappears from the home screen if a new screening is in progress, blocking access to recommendations.
5. Error Prevention	Error-tolerant systems and simplified recovery paths for motor-related interaction errors (Amouzadeh et al., 2025; Wildenbos et al., 2018).	**Logout:** No confirmation message appears before the logout action is finalized.
6. Recognition Rather than Recall	Visible main functions and flat hierarchies reduce memory demands (Liu et al., 2021; Morey et al., 2019).	**Mobile View:** Main functionalities are not immediately available on the homepage when accessed via a phone.

		“Read More”: These generic labels do not explicitly state what content follows.
		Lack of signifiers: Some elements are styled only in bold without additional signifiers to distinguish them from static headers.
		False affordances: Certain static elements are styled as buttons or cards, suggesting they are clickable when they are actually non-interactive.
		Persistent access: Recommendations and graph views are not persistently accessible when seeing results.
7. Flexibility & Efficiency of Use	Support for varied interaction methods (audio/visual) and adjustable interfaces (Liu et al., 2021; Amouzadeh et al., 2025; Harshini Ramdowar et al., 2023; McCarthy et al., 2024; Khamaj, 2024).	**Customization:** Absence of adjustable text sizes and audio input/output options. Reminders cannot be timed.
		Links: Recommended external sources might be too general to be relevant.
8. Aesthetic & Minimalist Design	Prioritize white space to reduce cognitive overload. Avoid scrolling to reduce interaction burden. Use large sans-serif fonts to enhance visual accessibility. Use high contrast to improve readability (Liu et al., 2021; Morey et al., 2019).	**Scrolling:** The interface requires extensive vertical scrolling, which increases the effort required to navigate content. However, changing this might mean requiring more clicks.
9. Help Users Recover from Errors	Recovery-oriented feedback using plain language and concise phrasing to support comprehension (Gomez-Hernandez et al.,	**Special Characters:** The system does not explain the meaning of this term when the user is creating a password.

	2023; Wildenbos et al., 2018).	
10. Help & Documentation	Step-by-step guidance, video tutorials, and concise onboarding support comprehension and are preferred over exploratory learning (Gomez-Hernandez et al., 2023; Wildenbos et al., 2018).	**Onboarding:** The tool lacks onboarding or step-by-step guidance.

9.3 Prototype Redesign

The following figures show some of the main changes implemented in the Next-of-Kin Compass prototype. Figure 7 shows the previous state of the results interface, while Figure 8 shows the change made.

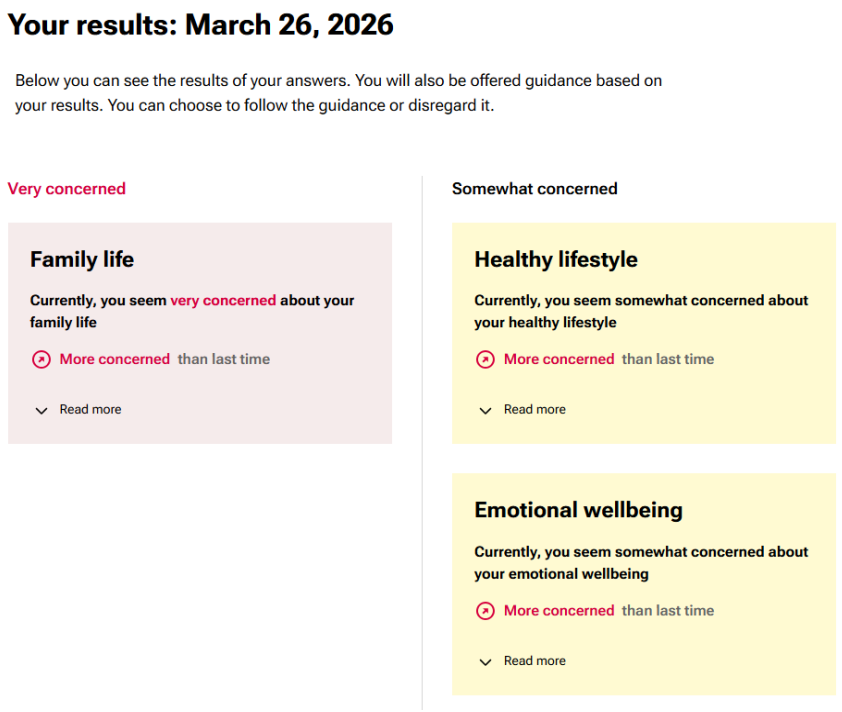


Figure 7. Read More labels. *The interface shows the results after a screening, with a "Read More" label for each of the categories.*

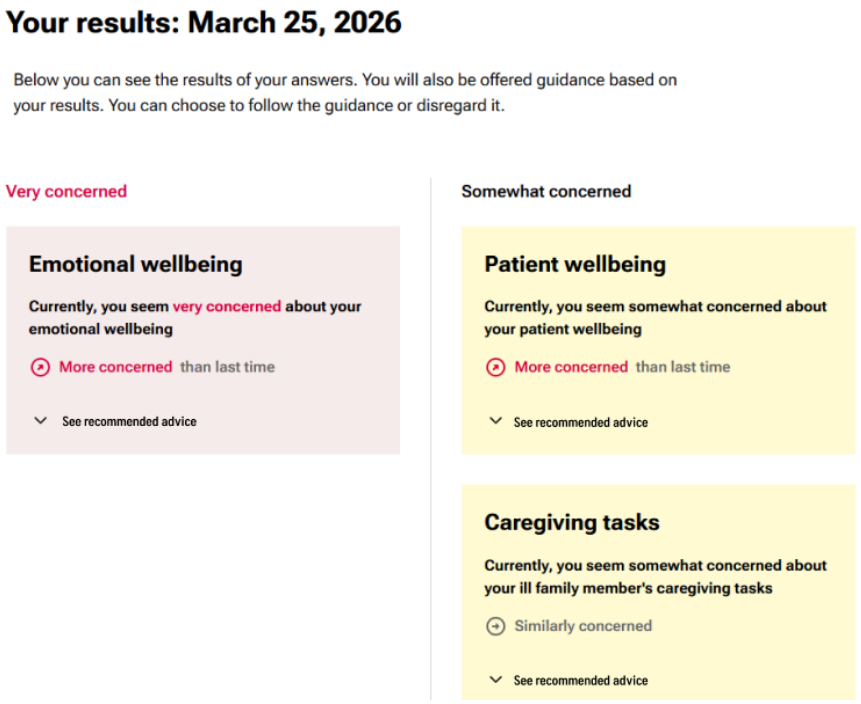


Figure 8. Descriptive labels replacing generic text. *Labels such as "See Advice" replace generic text, reducing cognitive load and improving predictability for older users.*

Additionally, Figure 9 shows the previous state of the recommendations interface, while Figure 10 shows the change made.

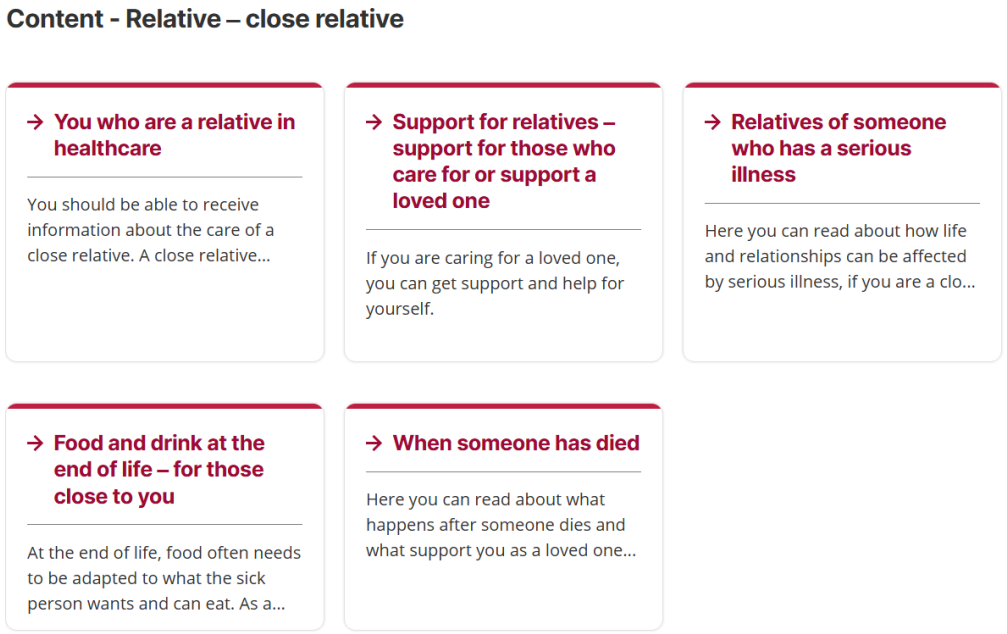


Figure 9. Generic advice. *The interface shows an example of the information recommended to the user to read.*

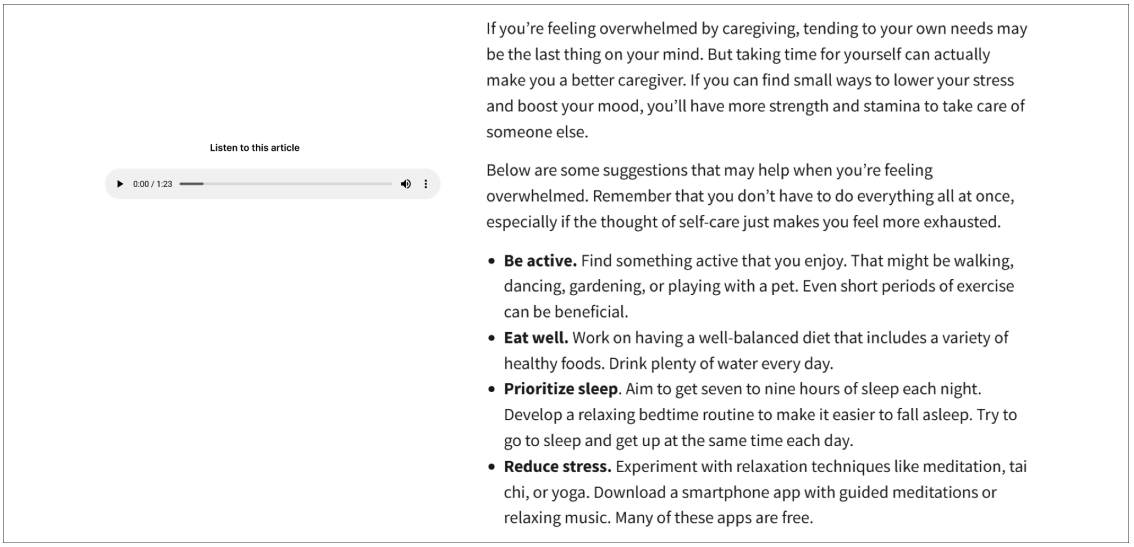


Figure 10. Personalised content based on screening results. *Helps users quickly find relevant information and reduces search effort.*

Next, Figure 11 shows the previous state of an interface with information for supporting the user, while Figure 12 shows the change made.



Figure 11. Without audio playback. *The interface shows an article recommended to read for users.*



Figure 12. Audio Playback Feature *Audio narration provides an alternative to reading long text, supporting users with visual impairments or limited time*

The previous state of the interface to create a new note is shown in Figure 13, while Figure 14 shows the change made.

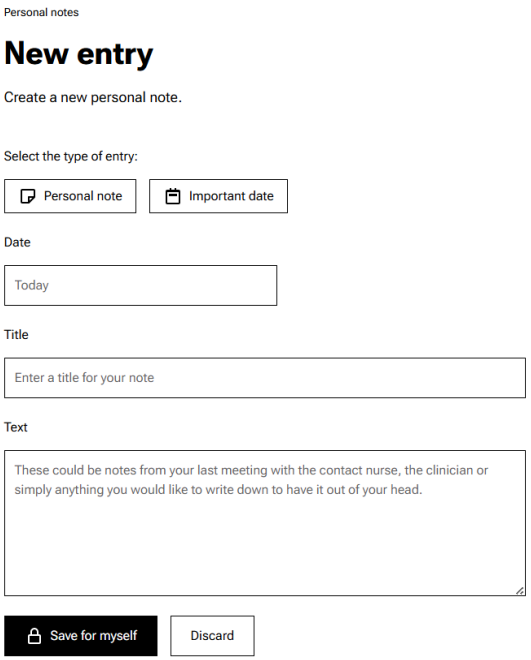


Figure 13. Without voice input. *The interface shows the form for creating a new entry in the personal notes of the application.*

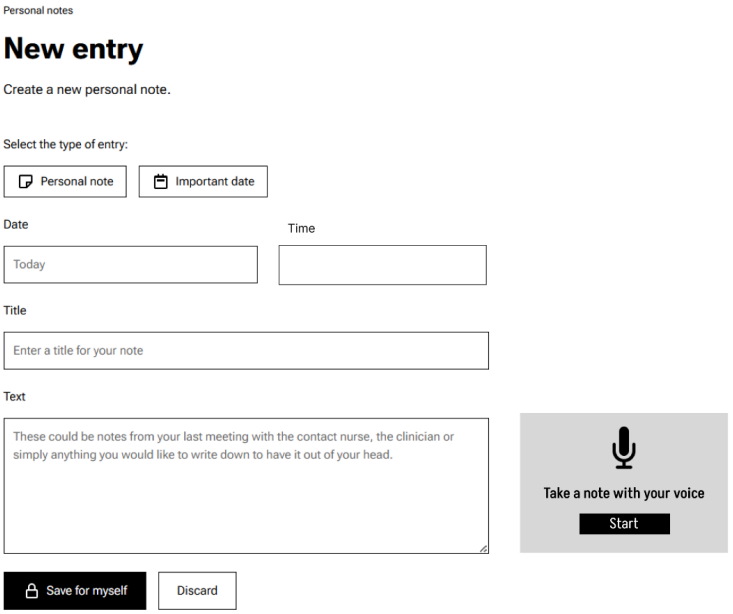


Figure 14. Voice-Note Input *Voice-note recording reduces typing effort and supports users with motor limitations.*

Finally, Figure 15 shows the introduction of the onboarding to the user when first using the application.

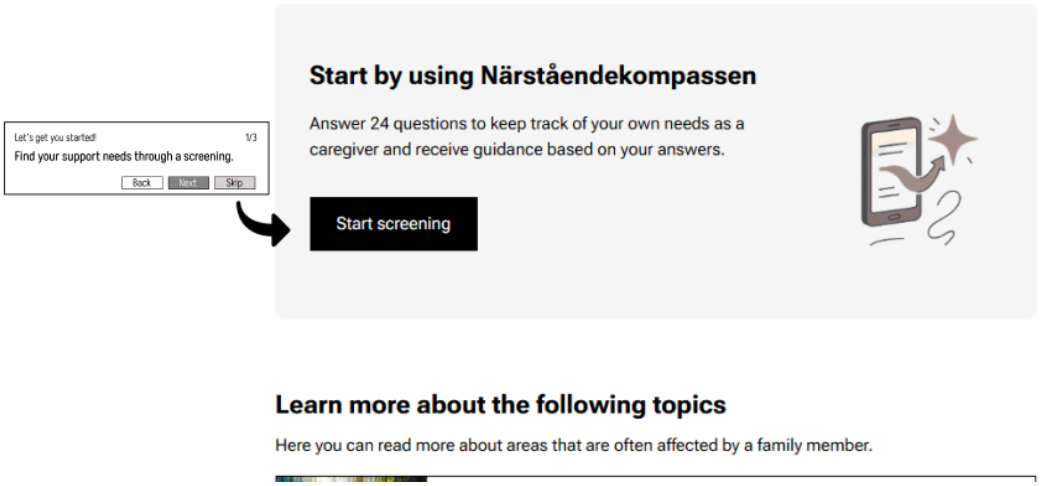


Figure 15. Onboarding Walkthrough *The onboarding flow introduces key features step-by-step, supporting users with low digital confidence.*

9.4 Professional Handout

Designing Digital Health Tools for Older Caregivers

Design Guidelines for designers, developers, and researchers in Human-Computer Interaction (HCI), digital health, and ageing studies

Digital engagement fails when user capacity is low and system complexity is high. Older family caregivers face a double burden: the challenges of ageing (reduced vision, slower processing, lower digital confidence) and the relentless demands of caregiving (time pressure, emotional strain, coordination overhead). Most digital health tools are designed for neither. These guidelines address both.

Core principle: Capacity–Demand Alignment
Design so interaction demands never exceed what a burdened and ageing older caregiver can give.

Professional Design Checklist

Design Guideline	Description	Example Design Strategies
Minimise cumulative cognitive load	Reduce cognitive burden across interaction sequences by lowering memory dependence, decision complexity, and interpretive effort.	Simplified navigation structures; progressive disclosure; persistent on-screen guidance; limiting the number of choices per screen.
Design for temporal scarcity	Support fragmented, intermittent use by reducing dependency on uninterrupted, sequential task completion.	Autosave functionality; resumable task flows; short interaction sequences; continuity-preserving reminders.
Contain emotional burden	Reduce interaction-induced emotional strain by ensuring predictable and non-escalating interaction flows.	Calm and neutral notification language; consistent screen layouts; minimised use of disruptive pop-ups or urgent alerts.
Support patient-centered perspectives	Reduce interaction cost by integrating caregiver-support functions directly into patient-centred tasks.	Stress-management prompts embedded in medication or appointment flows; contextual caregiver check-ins within care tasks.

Reduce system-level fragmentation	Decrease coordination burden by aligning with existing healthcare infrastructures and limiting cross-platform transitions.	Consolidated dashboards; consistent navigation patterns across services; integration with established systems such as 1177.
Support self-efficacy through interaction design	Enhance perceived control through clear, predictable, and transparent system behaviour.	Clear affordances and labels; consistent navigation structures; visible progress indicators; immediate confirmation feedback after actions.
Accommodate multimodal accessibility constraints	Address sensory, motor, and cognitive limitations through accessible and flexible interaction modalities.	Large touch targets; high-contrast visual design; adjustable text size; voice input; audio-supported instructions.

Key Takeaway

Older caregivers do not disengage because of age alone. They disengage when digital systems demand more cognitive, emotional, and coordination effort than they can realistically provide. Effective design reduces cumulative burden and not just isolated usability issues**.**

Original thesis: A Socio-Technical Approach to Designing Digital Health Tools for Older Caregivers: Insights from a Case Study of the Next-of-Kin Compass (2026) – full academic text available.